Reaction Intermediates and Advanced Carbocation Chemistry

Structure-based notes and problem workbook — WORK-IN-PROGRESS BUILD: Parts 0–4 and the opening of Part 7 (Parts 5, 6, rest of 7 onward not yet written)

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Part 0 Reaction Intermediates: Bond Fission, Intermediates, Electrophiles and Nucleophiles

0.1 Homolytic and heterolytic bond fission

A covalent A–B bond can break in two ways. In **homolysis** the bonding pair is shared equally: each fragment keeps one electron, giving two radicals. Single-electron movement is shown with **fishhook** (half-headed) arrows. In **heterolysis** the whole pair goes to one fragment, giving a cation and an anion; the electron pair movement is shown with a **full-headed** curved arrow that starts at the bond and ends at the atom that keeps the pair. Reaction arrows (→, ⇌) are never used for electron flow.

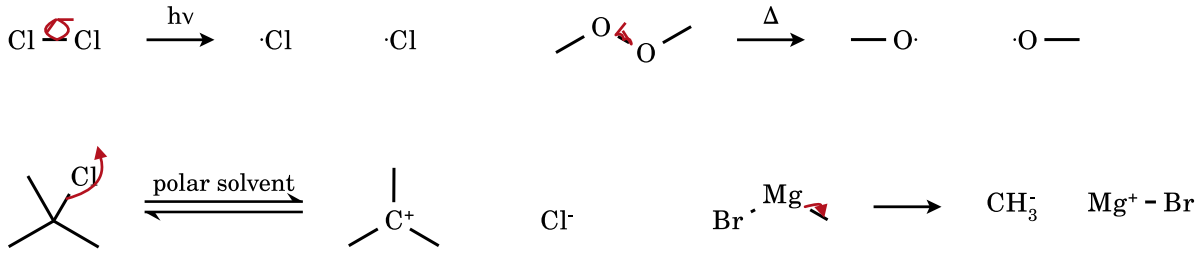


Figure 0.1 Homolysis (fishhook arrows; Cl_2 with light, a peroxide with heat) and heterolysis (full-headed arrows; an alkyl halide in a polar solvent, a C–metal bond).

- **Homolysis** is favoured by non-polar bonds (Cl–Cl, O–O, C–C, N=N of azo compounds), non-polar solvents or the gas phase, heat or UV light, and by radical-stabilising substituents on the fragments. Typical bond dissociation enthalpies (kcal mol^{-1}): Cl–Cl 58, Br–Br 46, I–I 36, HO–OH ≈ 51, RO–OR ≈ 38, CH_3–CH_3 90, H–CH_3 105, H–$C(CH_3)_3$ ≈ 96. Weaker bond ⇒ lower temperature needed.
- **Heterolysis** is favoured by polar bonds (large $\Delta\chi$), polar protic solvents that solvate both ions (hydrogen bonding to the anion, dipole/lone-pair interaction with the cation), a stable cation and a stable anion, and Lewis-acid assistance. In the gas phase heterolysis costs far more energy than homolysis; the solvation energy is what makes it feasible in solution.
- **Direction of heterolysis**: the electron pair stays with the more electronegative atom / the better leaving group (R–X → R^+ + X^-; R–MgBr → R^- + $MgBr^+$ with carbon taking the pair because C is more electronegative than Mg).
- **Typical initiators of homolysis**: peroxides (benzoyl peroxide, di-tert-butyl peroxide), azobisisobutyronitrile (AIBN, loses N_2 at 60–80 °C), halogens with light, N-bromosuccinimide with light/peroxide.

0.2 The reaction intermediates at a glance

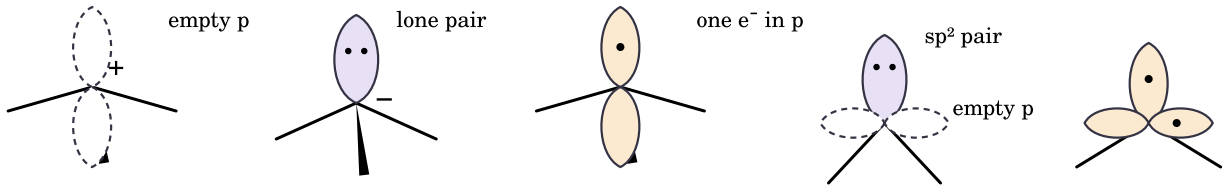


Figure 0.2 Orbital cartoons of the intermediates treated in this book (R = substituent). Solid lobes = occupied; dashed lobes = empty.

Intermediate	Valence e⁻ at the centre	Shape / hybridisation	Typical generation	Characteristic fate
Carbocation (carbenium)	6 (sextet)	planar, sp^2; linear sp for vinyl, acylium	heterolysis of R–X, ROH_2^+; H^+ + alkene; diazonium N_2 loss	nucleophile capture, E1 ($-H^+$), rearrangement, addition to π bonds
Carbanion	8 (3 bonds + lone pair)	pyramidal sp^3 (alkyl); planar when delocalised; sp (acetylide)	deprotonation; metal–halogen exchange; decarboxylation; Grignard/organolithium C–M	protonation, SN2, addition to C=O, E1cB, elimination
Free radical	7	planar (or shallow pyramid), sp^2; SOMO is a p orbital	homolysis (heat/light/initiator); single-electron transfer	chain propagation (abstraction/addition), recombination, disproportionation
Carbene	6 (2 bonds + 2 nonbonding e⁻)	singlet: bent sp^2, pair + empty p; triplet: bent/linear, two unpaired e⁻	α-elimination ($CHCl_3$/KOH); diazo or diazirine loss of N_2; carbenoid (Zn/Cu, CH_2I_2)	cyclopropanation, C–H insertion, ylide formation, Wolff rearrangement
Nitrene	6 (1 bond + 2 lone-pair-type e⁻ sets)	R–N: singlet or triplet (triplet usually ground state)	azide photolysis/pyrolysis; α-elimination from N-haloamides (concerted in Hofmann/Curtius)	insertion, aziridination, rearrangement, dimerisation
Benzyne (aryne)	ring C with in-plane 'third' bond	strained, planar; extra bond from sp^2 overlap in plane	ArX + $NaNH_2$/KNH_2; anthranilic acid + RONO; 2-(Me_3Si)C_6H_4OTf + F^-	nucleophilic addition, Diels–Alder, [2+2]
Radical ion	odd electron + charge	delocalised SOMO in π* (anion) or π (cation)	Na/NH_3 (Birch), Na + ketone/ nitrile, electrolysis, photo-induced ET	protonation, dimerisation, fragmentation, further e⁻ transfer

0.3 Electrophiles and nucleophiles (definitions)

A **nucleophile** is a species that donates an electron pair to form a new bond to an electron-poor atom (Lewis base character in a bond-forming step). An **electrophile** is a species that accepts an electron pair to form a new bond (Lewis acid character). In curved-arrow language, the arrow always leaves a nucleophile and arrives at an electrophile.

Basicity and acidity refer to proton transfer (equilibrium); nucleophilicity and electrophilicity refer to bond formation to atoms other than H (rate). Only the *definitions and classification* are treated here; no ranking of nucleophilicity is attempted.

Class	Nucleophiles	Electrophiles
Anions	HO^-, RO^-, X^-, CN^-, RS^-, N_3^-, R^- (carbanions, enolates), H^- (hydride donors)	—
Neutral with lone pair	H_2O, ROH, NH_3, RNH_2, R_3N, R_2S, R_3P, ROR	—
π systems	alkenes, alkynes, arenes, enols/enol ethers/enamines (C-nucleophiles)	C=C bearing EWG (Michael acceptors), C ≡ C–EWG, arenes with NO_2 (SNAr)
σ bonds	C–M (M = Li, Mg, Zn), B–H, Al–H, Si–C (allyl-/aryl-silanes)	polarised C–X (X = halogen, OTs) – carbon is electrophilic
Cations	—	H^+, R^+, ArN_2^+, NO_2^+, NO^+, X^+ (from X_2), Hg^{2+}, Ag^+, acylium RCO^+, iminium, oxocarbenium
Neutral, electron-deficient	—	BF_3, $AlCl_3$, SO_3, $FeBr_3$, $ZnCl_2$ (Lewis acids); carbenes (singlet); R–C=O, C=N, C ≡ N carbon; CO_2; epoxides; α,β-unsaturated carbonyls
Radicals	$•CR_2OH$, $•CH_2CH(OH)$… (nucleophilic, SOMO high)	$•CF_3$, $•CCl_3$, Br•, RC(=O)O•, $•CH(CO_2R)_2$ (electrophilic, SOMO low)

AMBIDENT NUCLEOPHILES (TWO REACTIVE SITES)

- CN^- (C or N), NO_2^- (N or O), SCN^- (S or N), enolates (C or O), phenoxide (O or ring C), thiourea/ thioamide (S or N). The site of attack depends on the electrophile and conditions (hard/soft, ion pairing, solvent); e.g. R–Br + KCN gives mainly R–CN, R–Br + AgCN gives mainly R–NC (isonitrile) because Ag^+ ties up the carbon end and pushes S_N1-like attack through N; enolate + R–X (soft) gives C-alkylation; enolate + R_3SiCl or RCOCl (hard) favours O.
- Ambident *electrophiles*: α,β-unsaturated carbonyls (C=O carbon, 1,2-attack; β-carbon, 1,4-attack), allyl halides (S_N2 vs S_N2'), epoxides, aldehyde/imine carbons.

Identification routine. (1) Draw lone pairs and π bonds; (2) mark atoms with δ^+ (attached to electronegative atoms) or an empty orbital; (3) join the most electron-rich site to the most electron-poor site; (4) check that every atom keeps a legal valence and the charges add up. Species such as carbenes are *ambiphilic* (singlet carbenes have a lone pair and an empty p orbital) and radicals can behave as either, according to the energy of the SOMO.

Part 1 Carbocations: Definitions, Classification, Geometry and Bonding

1.1 Definitions: carbocation, carbenium ion, carbonium ion

Carbocation is the general term for any species with a positively charged carbon atom. Two families are distinguished by the *coordination number* of that carbon:

- **Carbenium ions** (classical, tricoordinate): the positive carbon has **three** substituents and an empty p orbital, e.g. CH_3^+, $(CH_3)_3C^+$, $C_6H_5CH_2^+$. Formally they are derived from

a neutral carbene or an alkane by removal of H^-. These are the ions of ordinary organic mechanisms (S_N1, E1, electrophilic addition).

- **Carbonium ions** (non-classical, hypercoordinate): the positive carbon is **five- (or higher)** coordinate through a three-centre two-electron (3c–2e) bond, e.g. CH_5^+, $C_2H_7^+$; formally derived by protonation (or alkylation) of an alkane σ bond. The 2-norbornyl cation is a bridged (σ-delocalised) species of this bonding type.
- The older usage 'carbonium ion' for *every* carbocation (including t-Bu^+) is obsolete; the two names must not be interchanged. In this book 'carbocation' is the generic term, 'carbenium' is used when precision is needed, and 'carbonium' always means a hypercoordinate (3c–2e) ion.

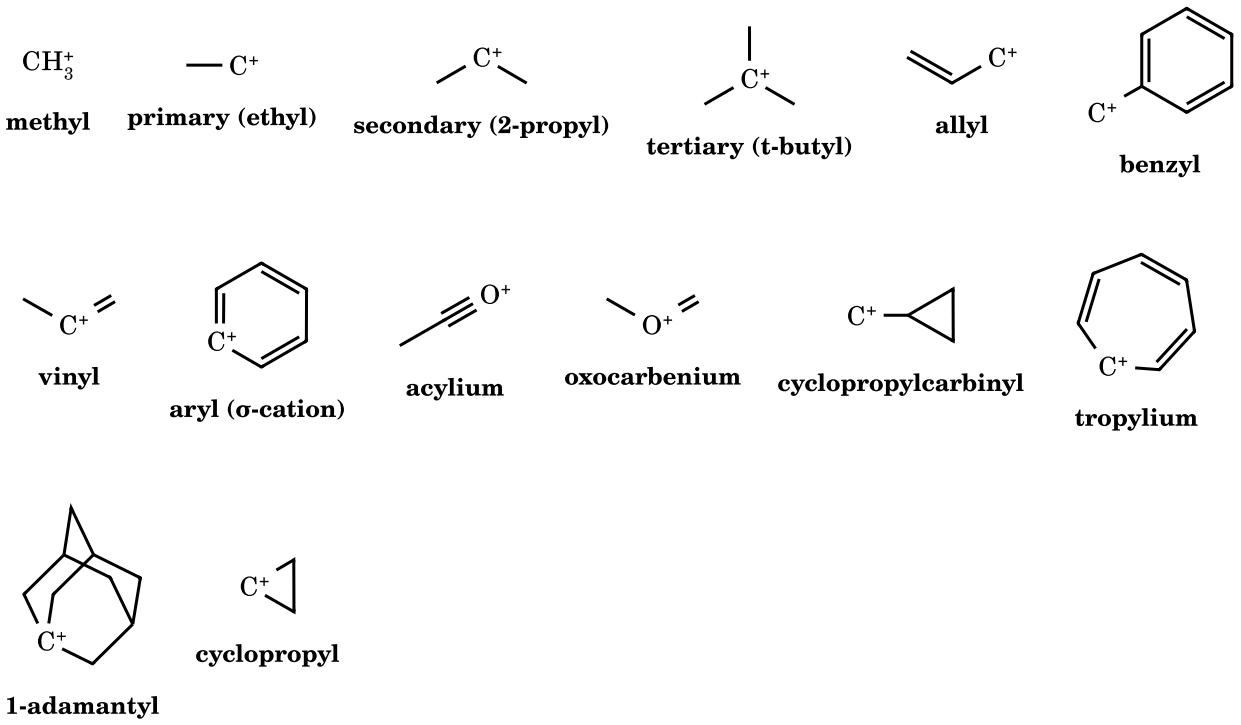


Figure 1.1 Gallery of carbocation types. The skeletal convention is used: hydrogens on carbon are not drawn; a symbol 'C⁺' is a cationic carbon with its implicit hydrogens (e.g. 'C⁺' in the ethyl cation is CH_2^+).

1.2 Geometry, hybridisation and the empty p orbital

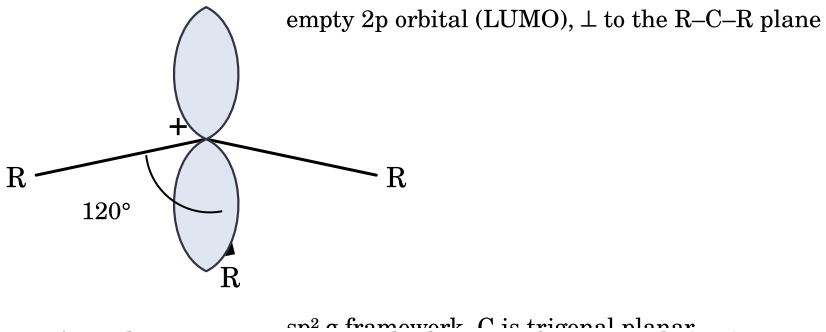


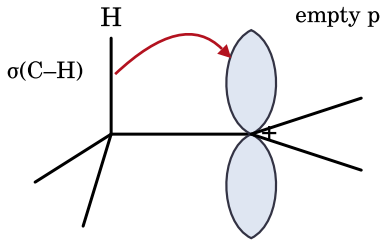
Figure 1.2 A carbenium ion: trigonal-planar σ framework (sp^2), empty 2p orbital perpendicular to the plane.

- **Planarity.** The three σ bonds use sp^2 hybrids (120°); the remaining p orbital is empty and is the LUMO. Planarity places the empty orbital in a position to be attacked from *either* face – the basis of racemisation in S_N1.

- **Bond lengths.** C^+–C bonds are shorter than normal $C(sp^3)$–$C(sp^3)$ bonds because of greater s character and hyperconjugative double-bond character.
- **Linear cations.** Vinyl ($R–C^+=CR_2$) and acylium ($R–C\equiv O^+$) cations have an sp carbon (180°); the vinyl cation's empty p orbital lies *in* the plane of the double bond, orthogonal to the π bond, so the π system cannot stabilise it by resonance. The phenyl (aryl) cation has its empty orbital in the plane of the ring, orthogonal to the aromatic π system – hence it is exceptionally unstable.
- **Pyramidalisation is penalised.** If the carbon is forced out of planarity (bridgehead of a small bicycle), overlap with substituent σ orbitals is poor and the p character is lost: 1-norbornyl (bicyclo[2.2.1]hept-1-yl) cations are very hard to form, whereas the larger, more flexible 1-adamantyl and bicyclo[2.2.2]oct-1-yl ions form far more easily.
- **Electrophilicity.** The empty orbital makes a carbocation a strong electrophile and Lewis acid: gas-phase hydride affinities (recalled, kcal mol^{-1}, approximate): CH_3^+ ≈ 312, i-Pr^+ ≈ 249, t-Bu^+ ≈ 231, benzyl ≈ 238, allyl ≈ 256 – the smaller the hydride affinity, the more stable the cation relative to its hydrocarbon. In solution Mayr's E parameter ranks benzhydrylium ions from E ≈ −7 (strongly donor-substituted) to E ≈ +6 (Ph_2CH^+).

1.3 Orbital overlap requirements and delocalisation of charge

A cation is stabilised to the extent that electron density from neighbouring orbitals can flow into its empty p orbital. Three kinds of donor exist and each has a geometric requirement: (i) **lone pairs** (n → p) on an attached heteroatom (OR, NR_2, SR, halogen): the lone-pair orbital must be parallel to the empty p orbital; (ii) **π bonds** (π → p) in an allyl, benzyl or vinyl system: the π orbitals must be parallel (coplanar σ frameworks); (iii) **σ bonds** ($\sigma_{C–H}$, $\sigma_{C–C}$ → p) – hyperconjugation: the donor σ bond must be close to parallel to the empty p orbital, i.e. roughly perpendicular to the plane of the cation (dihedral angle near 0°; at 90° the overlap is zero).

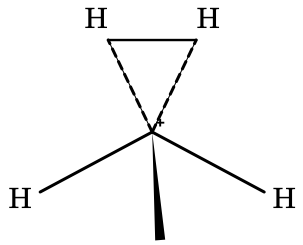


C–H parallel to the p axis: maximum overlap

Figure 1.3 Hyperconjugation: the neighbouring σ(C–H) bond donates into the empty p orbital when the two are parallel.

Localised versus delocalised charge. In t-Bu^+ the charge is formally on one carbon but is shared with nine hydrogens by hyperconjugation; in allyl and benzyl cations resonance places partial positive charge on two or more carbons (shown with double-headed arrows between contributors, never with reaction arrows); in σ-bridged non-classical ions (2-norbornyl, cyclopropylcarbinyl, phenonium) the charge is spread over three or more carbons through *partial bonds* and no single Lewis structure is adequate. Resonance contributors differ only in electron positions; the nuclei do not move. If atoms move, the species are different (isomers, rapidly equilibrating or bridged).

1.4 Hypercoordinate (carbonium) ions



$[CH_5]^+$ — carbonium ion

Figure 1.4 CH_5^+: three ordinary C–H bonds and a three-centre two-electron C···H_2 unit (H_2 bound side-on to a CH_3^+ fragment).

Protonation of methane in a superacid or in a mass spectrometer ion source forms CH_5^+. High-level calculations and gas-phase infrared spectra show a structure with a CH_3^+ unit and a bound H_2, but with extremely facile exchange of all five hydrogens: it is a fluxional, hypercoordinate species with only *eight* valence electrons for five bonds. A carbonium ion is therefore **not** simply a 'more substituted' or 'more stable' carbenium ion; its 3c–2e bonding is a different bonding type. Their stability cannot be placed on the same 3° > 2° > 1° scale. Their usual fate is loss of H_2 (giving a carbenium ion; Part 2.E6), loss of a hydrocarbon fragment, or rapid proton scrambling.

1.5 Ion pairs, solvent and counterion

Ionisation passes through **contact (intimate) ion pairs** R^+X^-, then **solvent-separated ion pairs** $R^+\|X^-$, then free ions (Winstein). Nucleophile capture from the contact pair is stereo-selective (the counterion shields one face), from the solvent-separated pair largely statistical. The solvent ionises the substrate by (a) hydrogen bonding to the leaving group (protic solvents, TFE, HFIP are best) and (b) a high dielectric constant that screens charge; the nucleophilicity of the solvent matters much less in S_N1 than in S_N2. A **non-nucleophilic, weakly coordinating counterion** (SbF_6^-, $B(C_6F_5)_4^-$, carboranes, $Al(OC(CF_3)_3)_4^-$) and a non-basic solvent (SO_2ClF, SO_2, CH_2Cl_2) allow even tertiary cations to be observed (Olah, 1960s onward).

> **Scope and honesty note.** This book's definitions follow current IUPAC practice and standard physical-organic-chemistry usage. The IUPAC Gold Book, the textbooks listed in Part 17 and the archive.org page requested were not accessible in this authoring environment; where a numerical value is quoted from memory it is marked 'recalled' or given as an order only. See the Source Map for what was and was not consulted.

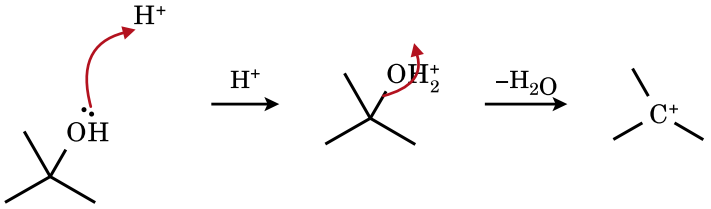
Part 2 Generation of Carbocations

Each of the following mechanisms was checked by computer: the curved arrows were applied to the drawn reactant graph and the result compared with the drawn product (valence, charge, atom conservation). Examples are grouped by generation route: 2.A alcohols, 2.B alkyl halides and sulfonates, 2.C alkenes/dienes/alkynes, 2.D solvolysis, 2.E other routes, 2.F amines with nitrous acid, 2.G aromatic compounds (arenium ions).

2.A Carbocations from alcohols

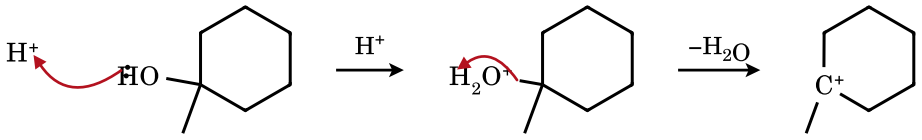
Alcohols do not ionise directly because HO^- is a poor leaving group (conjugate acid H_2O, $pK_a \approx 15.7$). Protonation, or coordination to a Lewis acid, converts it into H_2O (pK_a of $H_3O^+ \approx -1.7$) and ionisation then becomes possible *only if* the cation is stabilised enough: tertiary, benzylic, allylic or α-heteroatom-substituted. Primary and unactivated secondary alcohols generally react by concerted routes (E2, S_N2, or ionisation with simultaneous 1,2-shift) and are drawn that way below. Every arrow starts at an electron pair (lone pair or bond) and ends where that pair finishes.

2.A1 tert-Butyl alcohol, dilute strong acid



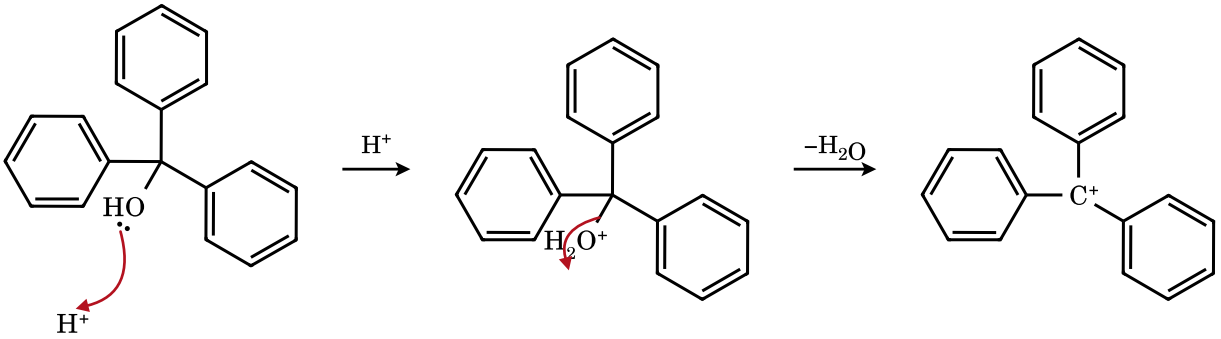
Rate-limiting loss of water from the oxonium ion gives the planar tert-butyl cation (nine hyperconjugating C–H bonds). In water the cation is captured faster than it eliminates unless the mixture is heated.

2.A2 1-Methylcyclohexanol, H_2SO_4



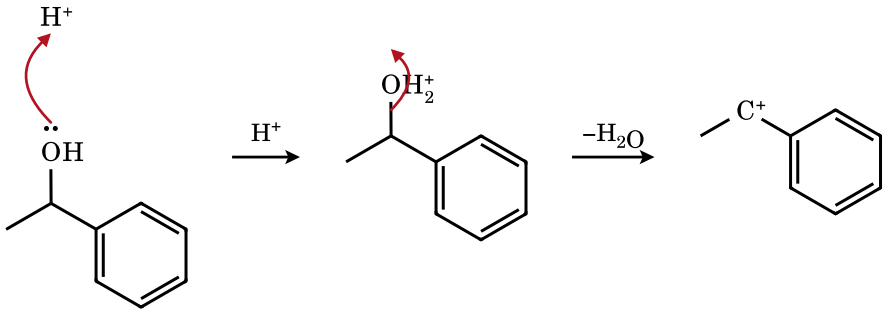
A tertiary ring cation: the cationic carbon is sp^2. E1 loss of H^+ competes and gives 1-methylcyclohexene (the Zaitsev alkene) on warming.

2.A3 Triphenylmethanol, conc. H_2SO_4



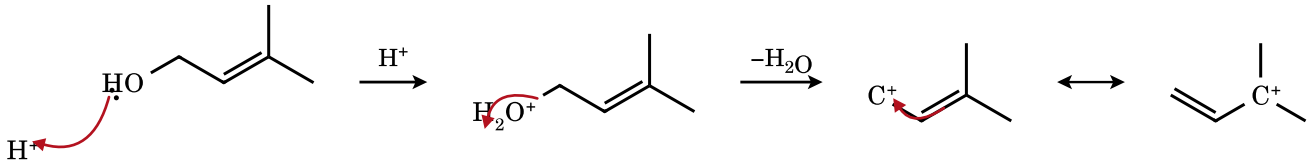
The yellow trityl cation is delocalised into three rings, which twist about 30° out of the central plane (propeller) because ortho-H···ortho-H contacts forbid full planarity. Isolable as BF_4^- or ClO_4^- salts.

2.A4 1-Phenylethanol, HBr (conc.)



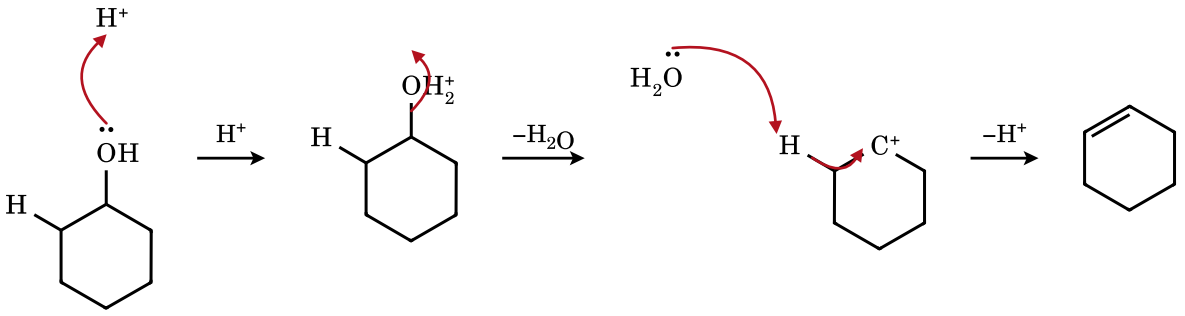
A secondary benzylic cation, stabilised by +M delocalisation into the ring (charge at o- and p-carbons). Bromide then captures it (S_N1).

2.A5 Prenol (3-methylbut-2-en-1-ol), H^+



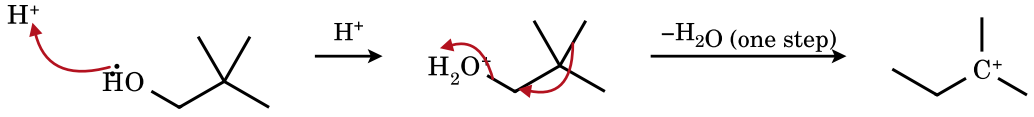
Ionisation puts the charge at C-1 (primary) *and*, by the double-headed resonance arrow, at C-3 (tertiary): one allyl cation, two contributors, with most of the positive charge on the tertiary terminus. Nucleophiles can attack at either end (S_N1 and S_N1').

2.A6 Cyclohexanol, 85 % H_3PO_4, heat (E1)



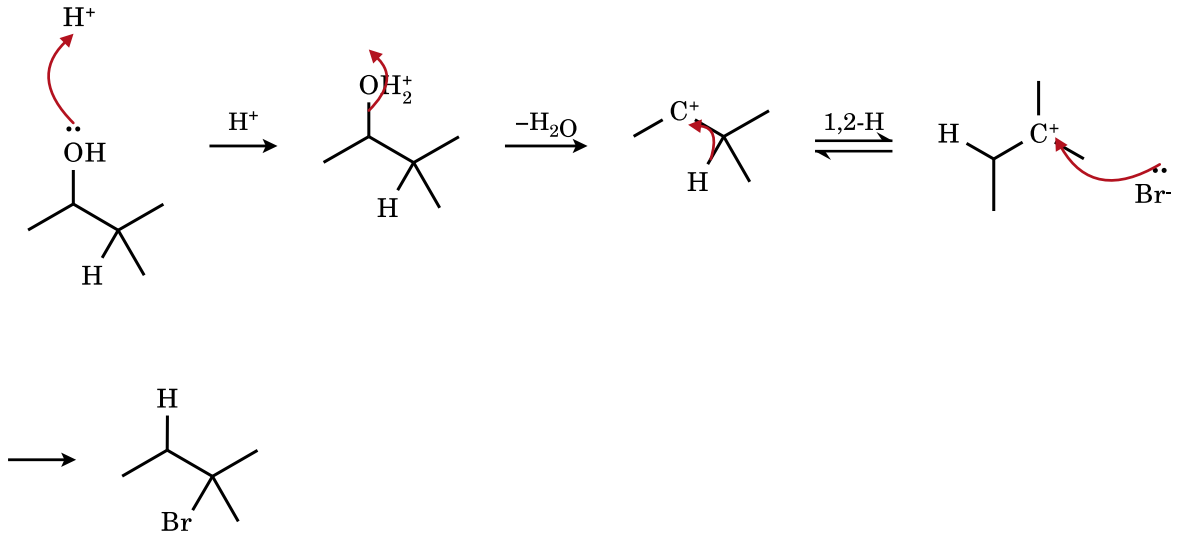
A secondary cation is formed only at elevated temperature and in poorly nucleophilic acid (H_3PO_4, HSO_4^-). The base removes a β-H whose C–H bond overlaps the empty p orbital; the C–H pair becomes the new π bond (E1). H_3O^+ regenerates the catalyst.

2.A7 Neopentyl alcohol, HBr, heat: ionisation with concerted CH_3 shift



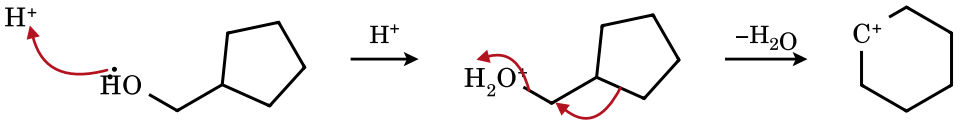
A free primary cation $(CH_3)_3CCH_2^+$ is not an intermediate: the C–CH_3 σ bond anti to the departing water pushes in as the leaving group departs, so the tertiary tert-amyl cation is reached directly (anchimeric σ-assistance). The same applies to neopentyl halides and tosylates.

2.A8 3-Methylbutan-2-ol, conc. HBr



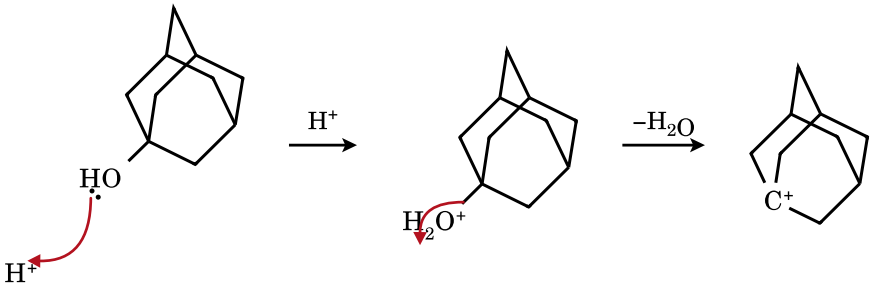
The secondary cation is formed with (or just before) the hydride shift: C–H (σ) → empty p. The tertiary cation is lower in energy and is captured by Br^-: the major product is 2-bromo-2-methylbutane, not 2-bromo-3-methylbutane.

2.A9 Cyclopentylmethanol, H^+: ionisation with ring expansion



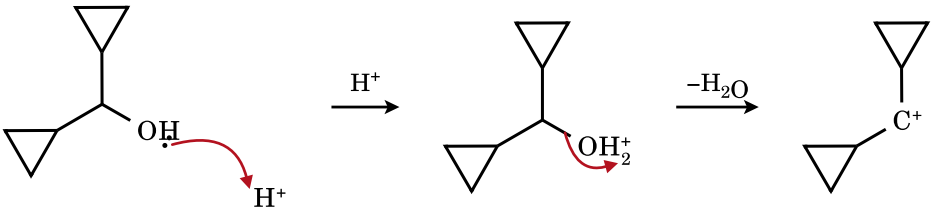
The ring C–C bond migrates to the departing CH_2 as water leaves, so the ring grows from five to six atoms and a secondary cation results; no primary cation intervenes. Ring strain relief is small (cyclopentane → cyclohexane) – the driving force is mainly 1° → 2°.

2.A10 1-Adamantanol, H_2SO_4 (bridgehead cation)



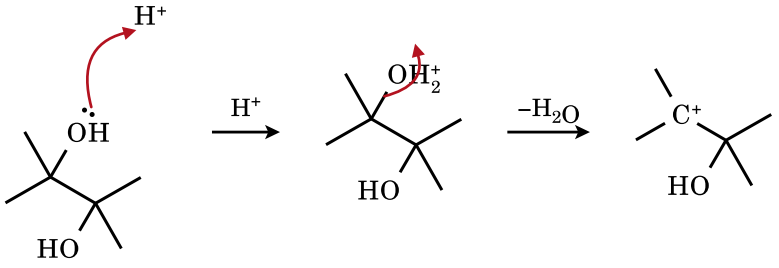
The 1-adamantyl cation is a tertiary bridgehead cation that forms readily: the cage is rigid, strain-free in the cation and the three C–C bonds at C-1 are held in good overlap with the empty orbital ($\sigma_{C–C}$ hyperconjugation). Contrast with bicyclo[2.2.1]hept-1-yl, where bridgehead cations are strongly disfavoured.

2.A11 Dicyclopropylmethanol, H^+



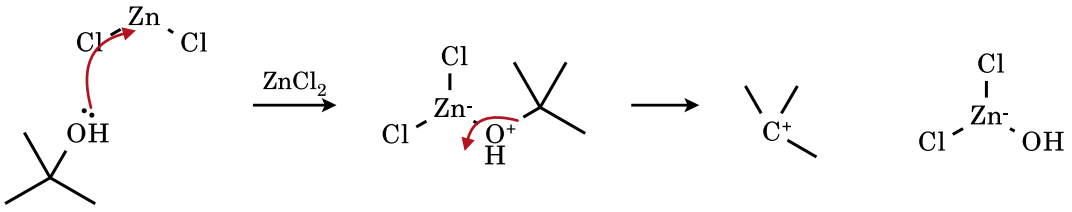
Each cyclopropyl ring donates its bent (Walsh-type) C–C σ bonds into the empty p orbital, which must lie in the *bisected* geometry. Dicyclopropylcarbinyl cations are far more stable than isopropyl- or even benzyl-type cations of the same substitution (see 5.5).

2.A12 Pinacol, 2,3-dimethylbutane-2,3-diol, H_2SO_4 (first step of the pinacol rearrangement)



The β-hydroxy tertiary cation is formed; its OH group is only inductively withdrawing and cannot stabilise it by resonance. A 1,2-methyl shift (Part 7) converts it into the oxygen-stabilised cation, which is the driving force.

2.A13 tert-Butyl alcohol, $ZnCl_2$/HCl (Lucas reagent), Lewis-acid ionisation

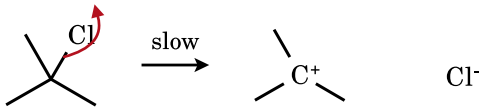


Zn(II) binds the oxygen lone pair; the C–O bond then ionises to the cation and the zincate $[HOZnCl_2]^-$. Tertiary alcohols give turbidity in seconds, secondary in minutes, primary not at all – the Lucas test is an S_N1 test.

2.B Carbocations from alkyl halides and sulfonates (ionisation)

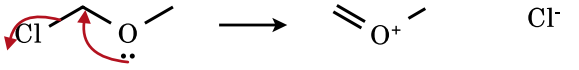
Heterolysis of a C–X bond gives $R^+\ X^-$; in solution the first species is an *ion pair* (contact, then solvent-separated) and only later free ions. Ionisation is helped by (i) a good leaving group ($I^- \approx Br^- > Cl^- \gg F^-$; sulfonates better still), (ii) a strongly ionising, hydrogen-bond-donating solvent, (iii) a halophilic Lewis acid (Ag^+, $AlCl_3$, SbF_5) and (iv) a cation stabilised by hyperconjugation, resonance or neighbouring-group bridging. Each example below is checked atom-by-atom: arrows start at the electron pair that moves and end where it comes to rest.

2.B1 tert-Butyl chloride in water/ethanol: ion pair



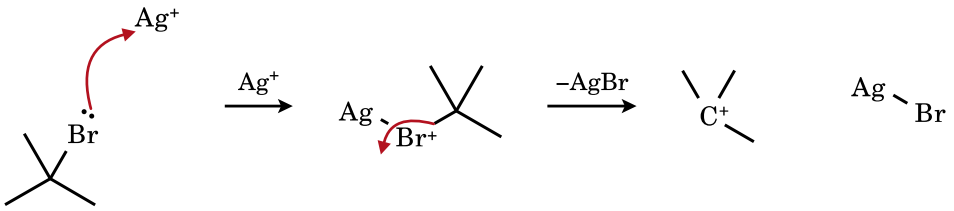
The rate-determining step: the σ(C–Cl) pair leaves with chlorine. In a solvent of high dielectric constant and H-bond donor ability (water, TFE, HCO_2H) the ion pair is stabilised and ionisation is faster by roughly five orders of magnitude than in ethanol (the Grunwald–Winstein Y scale is defined on exactly this substrate).

2.B9 Chloromethyl methyl ether (MOM-Cl)



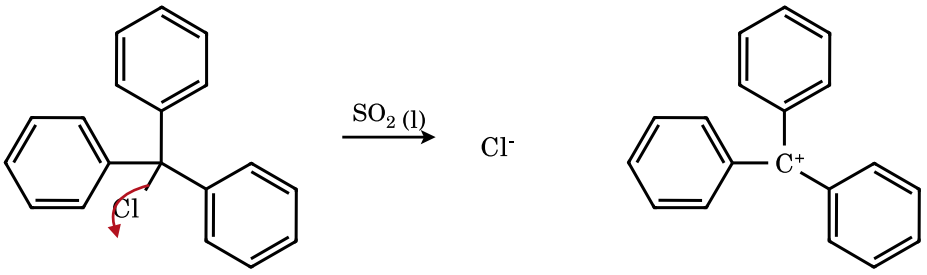
An α-heteroatom lone pair pushes out the halide: the 'carbocation' is really an oxocarbenium ion $CH_2{=}O^+CH_3$ in which every atom has an octet. α-Chloroethers ionise many orders of magnitude faster than simple alkyl chlorides of the same substitution.

2.B2 tert-Butyl bromide + $AgNO_3$ (Ag^+ assistance)



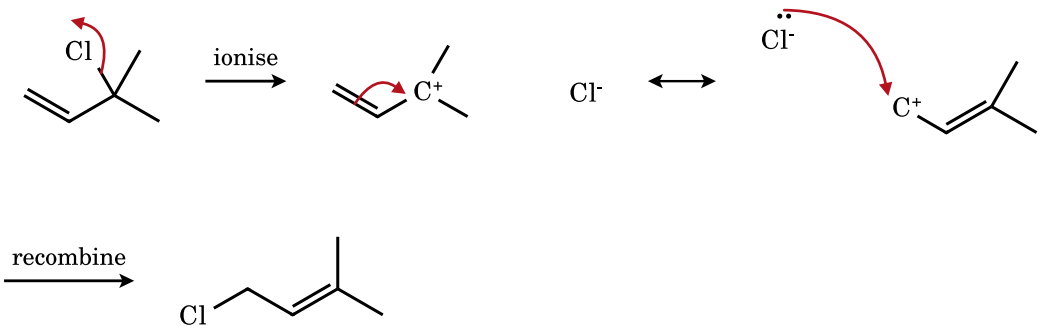
Ag^+ accepts a bromine lone pair (σ-complex), so the leaving group is effectively AgBr, which precipitates. This is why silver ion makes even primary neopentyl and secondary halides ionise – and rearrange.

2.B3 Trityl chloride in liquid SO_2



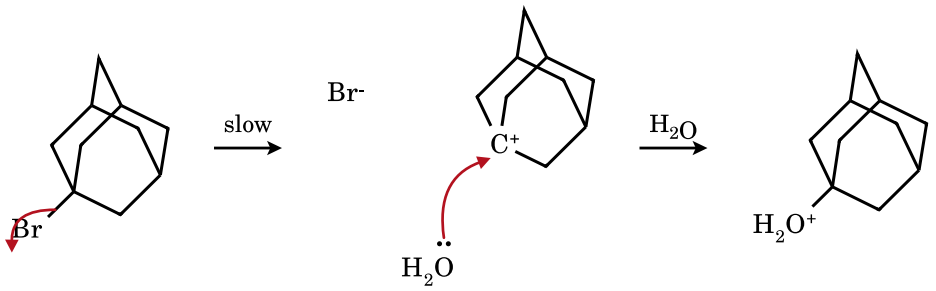
Ph_3CCl is a covalent solid but ionises in SO_2 (conducting yellow solution). The cation is stabilised by three aryl groups; compare Ph_2CHCl (ionises in the same solvents much more slowly) and $PhCH_2Cl$ (barely).

2.B4 3-Chloro-3-methylbut-1-ene ⇌ prenyl chloride: ion pair and allylic transposition



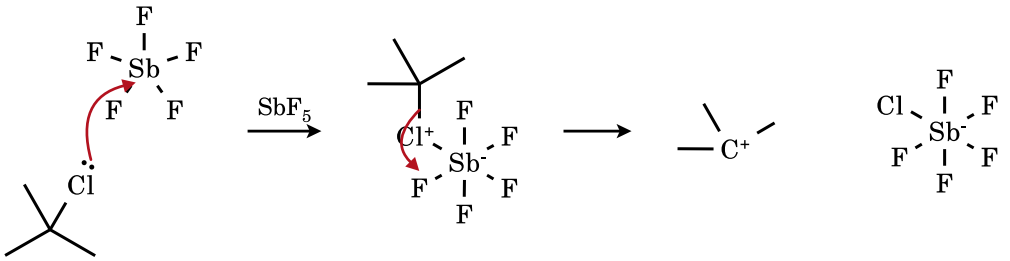
The same allyl cation is formed from either isomer. Chloride (kept nearby as an ion pair) returns to the less hindered primary terminus, giving the more substituted, thermodynamically more stable alkene (1,3-shift, S_Ni' / S_N1').

2.B5 1-Bromoadamantane in 80 % aqueous ethanol (bridgehead S_N1)



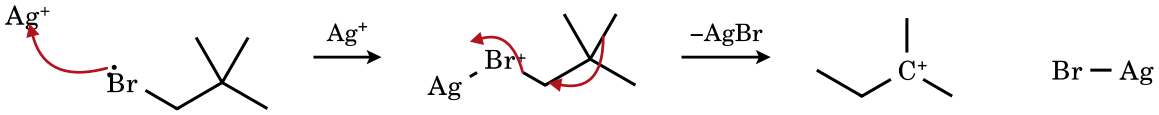
S_N2 is impossible at a bridgehead (no backside access) yet 1-adamantyl bromide solvolyses readily, because the adamantyl cage is nearly strain-free in the planar-ish cation. 1-Bromobicyclo[2.2.1]heptane, in contrast, is essentially inert (see 5.11).

2.B6 tert-Butyl chloride + SbF_5 in SO_2ClF, –78 °C (Olah)



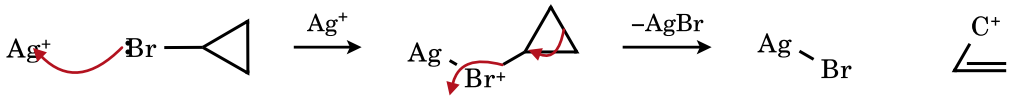
With a very strong halide acceptor and a non-nucleophilic medium the ion pair persists indefinitely: $(CH_3)_3C^+$ SbF_5Cl^- was observed by NMR (a single ^1H signal, the nine hydrogens equivalent by rapid methyl rotation), proving that a tertiary alkyl cation can be a real, long-lived species.

2.B7 Neopentyl bromide + $AgNO_3$ (concerted ionisation–methyl shift)



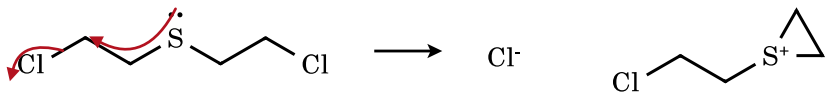
Neopentyl halides cannot undergo S_N2 (β-branching blocks backside attack) nor form a primary cation: when ionisation is forced, the anti-periplanar C–CH_3 bond migrates as bromide leaves.

2.B8 Cyclopropyl bromide + Ag^+: ionisation with disrotatory ring opening



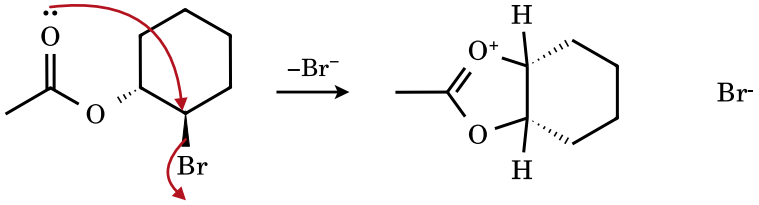
A cyclopropyl cation (sp^2 carbon in a 60° ring) is so strained that it is never formed as such: the C–C bond opposite the leaving group opens as the halide departs, giving the allyl cation (disrotatory, Woodward–Hoffmann, 2π electrons). Cyclopropyl halides solvolyse far more slowly than comparable secondary halides, and only with the assistance of the ring-opening.

2.B10 Sulfur mustard, $ClCH_2CH_2SCH_2CH_2Cl$: neighbouring-group participation



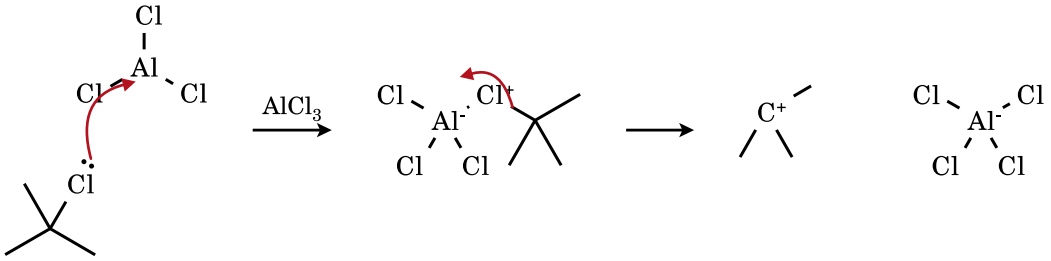
The β-sulfur displaces chloride intramolecularly (anchimeric assistance) to give a cyclic episulfonium (thiiranium) ion – a bridged cation in which charge sits mainly on sulfur and no open primary cation exists. It is a potent alkylating agent.

2.B11 trans-2-Bromocyclohexyl acetate: acetoxonium ion



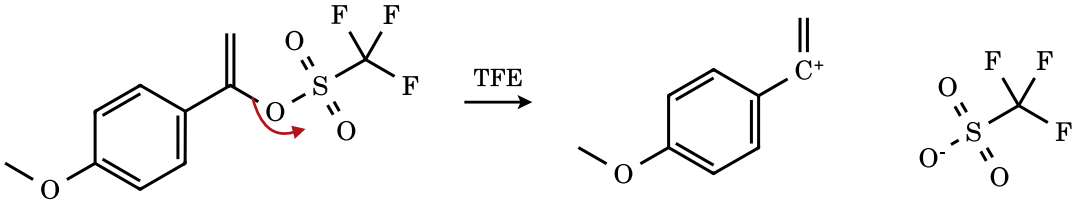
Only the *trans* (diaxial-capable) isomer reacts fast: the carbonyl oxygen attacks C-2 from the side opposite to Br, giving a 1,3-dioxolan-2-ylium (acetoxonium) ion; positive charge is shared by both oxygens and the carbon. Opening by a nucleophile at C-2 gives *net retention* (two inversions).

2.B12 tert-Butyl chloride + $AlCl_3$ (Friedel–Crafts alkylation)



$AlCl_3$ abstracts chloride to give t-Bu^+ $AlCl_4^-$. With primary halides the complex R–Cl→$AlCl_3$ is itself the electrophile and may rearrange (see 2.E4).

2.B13 1-(4-Methoxyphenyl)vinyl triflate in CF_3CH_2OH: vinyl cation



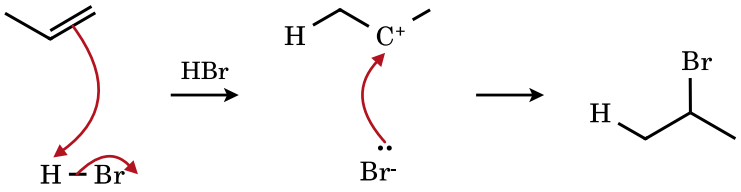
Vinyl (alkenyl) cations are real, linear sp-hybridised species but are high in energy; they form only from excellent leaving groups (triflate) and require stabilisation by an α-aryl (here a p-anisyl +M donor) or an α-alkyl group. The empty p orbital lies in the plane of the alkene, orthogonal to the π system.

2.C Carbocations from alkenes, dienes and alkynes (electrophilic addition)

A π bond is a nucleophile. Its electron pair attacks the electrophile (H^+, a carbocation, X^+, Hg^{2+}) and forms a σ bond to *one* carbon; the other carbon is left electron-deficient. The proton adds so that the **more stable** cation (Markovnikov) is produced, and the arrow

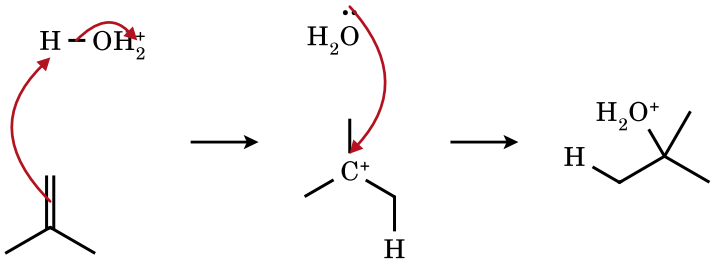
always runs from the π bond to the electrophile. When the electrophile has a lone pair that can reach back (Br, S, Hg), a **bridged onium ion** – not an open carbocation – is the intermediate, which is why no rearrangement and *anti* stereochemistry result.

2.C1 Propene + HBr



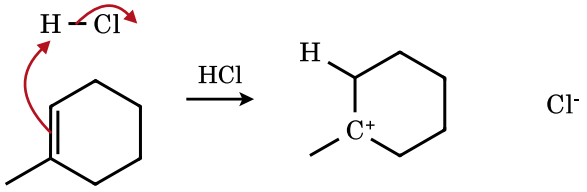
Protonation at C-3 puts the cation on the secondary C-2 (Markovnikov). The rate-limiting step is H^+ transfer from HBr to the alkene.

2.C2 2-Methylpropene + H_3O^+ (acid hydration)



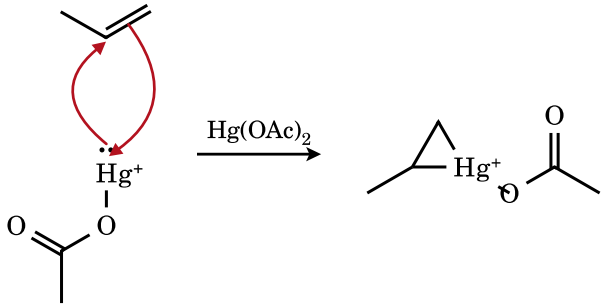
Only the tertiary cation is formed (the primary isobutyl cation would be ≈ 15 kcal/mol higher in energy). Capture by water and loss of H^+ gives tert-butyl alcohol; hydration is reversible (E1 reverse).

2.C3 1-Methylcyclohexene + HCl



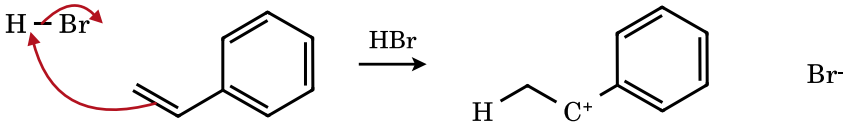
The proton adds to the less substituted alkene carbon, giving the tertiary ring cation. Capture by chloride from the open face, from either side, gives 1-chloro-1-methylcyclohexane (no stereocentre is created).

2.C7 Propene + $Hg(OAc)^+$ (oxymercuration)



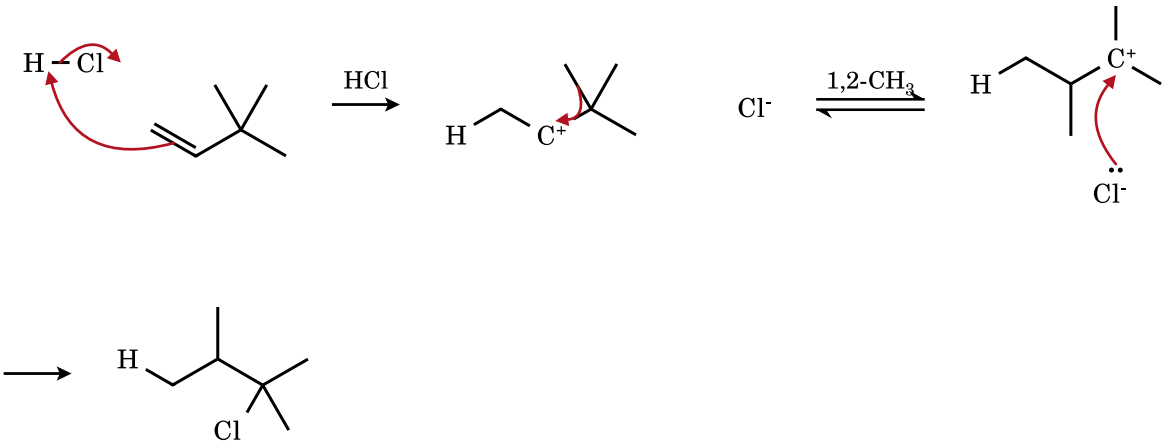
The mercurinium ion is bridged and delocalised; the nucleophile (H_2O) attacks the more substituted carbon (greater positive charge) with *anti* stereochemistry and, because no free cation forms, **without rearrangement**. $NaBH_4$ then replaces Hg by H.

2.C4 Styrene + HBr



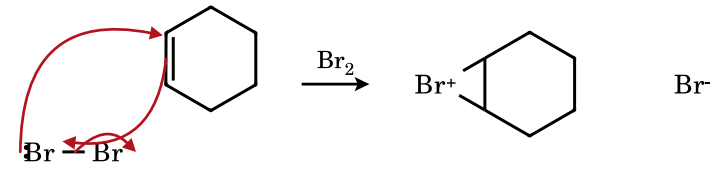
H^+ adds to the terminal CH_2 so that the cation is benzylic. The rate of addition to ring-substituted styrenes follows σ^+ ($\rho \approx -3$): a +M group in the para position speeds up the reaction.

2.C5 3,3-Dimethylbut-1-ene + HCl: addition with rearrangement



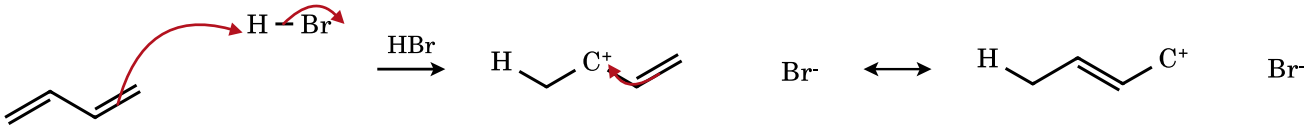
The secondary cation (neopentyl-type, with an adjacent quaternary carbon) is trapped in part by Cl^- (2-chloro-3,3-dimethylbutane, minor) but mostly undergoes a methyl shift to the tertiary cation: 2-chloro-2,3-dimethylbutane is the major product. A classic proof that alkene hydrohalogenation involves a cation.

2.C6 Cyclohexene + Br_2: bromonium ion



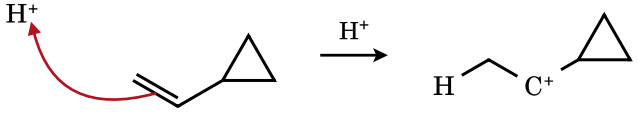
A bromine lone pair closes onto the carbon that lost the π electrons: the intermediate is a three-membered bromonium ion, not an open carbocation. Bromide then opens it from the face opposite to the ring (anti addition, trans-1,2-dibromocyclohexane).

2.C8 Buta-1,3-diene + HBr: allylic cation



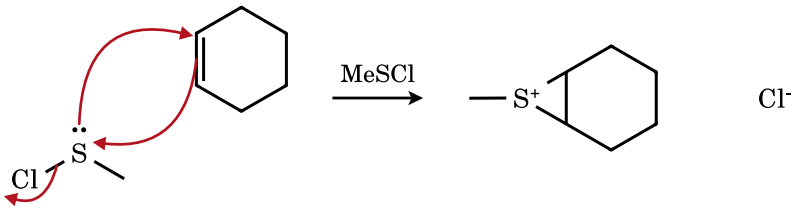
Protonation at the terminal carbon is preferred because the resulting cation is allylic (two contributors). Br^- can attack C-2 (1,2-adduct, formed faster at 0 °C – kinetic) or C-4 (1,4-adduct, more stable – thermodynamic); see Part 9.

2.C9 Vinylcyclopropane + H^+



A cyclopropylcarbinyl cation results, strongly stabilised by the neighbouring C–C σ bonds of the ring (bisected conformation, Part 5.5): vinylcyclopropanes protonate faster than simple 1,1-disubstituted alkenes.

2.C10 Cyclohexene + MeSCl: thiiranium ion



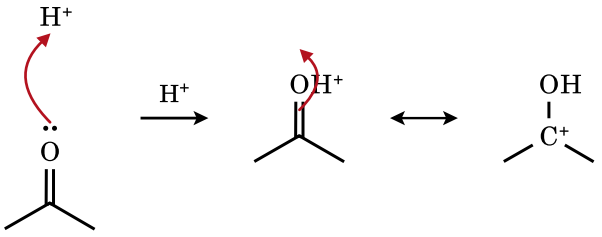
Sulfur's lone pair bridges the two carbons exactly as bromine does. Episulfonium ions give *anti*-addition products and can open at either carbon (Markovnikov or anti-Markovnikov, depending on substitution and solvent).

2.C11 Propyne + HBr: vinyl cation



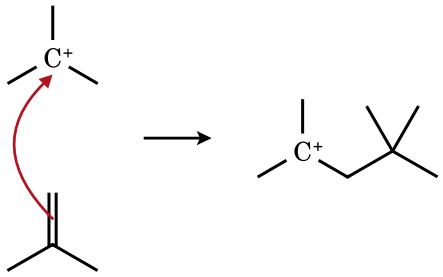
A secondary vinyl cation (linear, empty p in the alkene plane) is much less stable than a secondary alkyl cation, so alkynes are protonated more slowly than alkenes; with HX the addition is often termolecular (AdE3), avoiding a free vinyl cation.

2.C12 Acetone + H^+: oxocarbenium (protonated carbonyl)



A protonated carbonyl is a hydroxy-carbenium ion – a tertiary cation stabilised by the +M effect of OH. Resonance structure (right) shows why the carbon becomes electrophilic; acid-catalysed additions to C=O all start here.

2.C13 2-Methylpropene + t-Bu^+: C–C bond formation (cationic dimerisation)

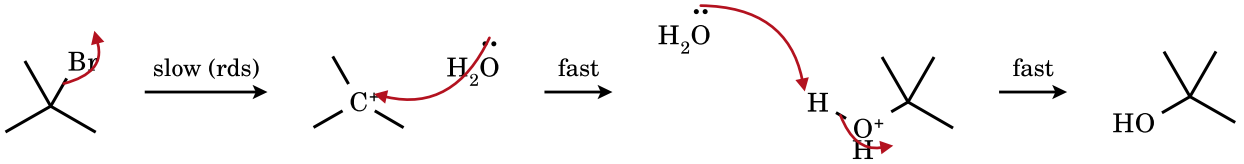


A carbocation is itself an electrophile for alkenes. The new tertiary cation can lose H^+ to give the diisobutylene isomers (2,4,4-trimethylpent-1-ene and -2-ene) or add a further isobutene (oligomerisation, polyisobutylene).

2.D Solvolysis and S_N1 substitution

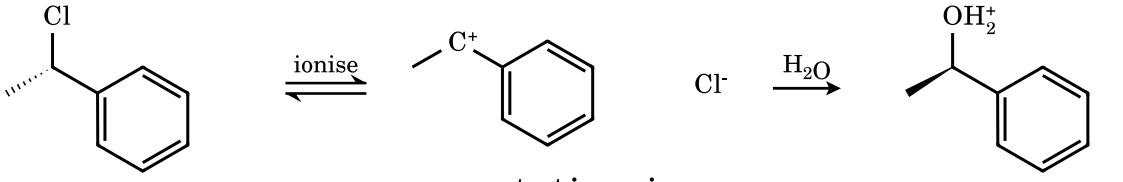
In solvolysis the solvent is both the ionising medium and the nucleophile. The kinetics are first order in substrate (rate = k[RX], independent of added nucleophile), and the stereochemical and product evidence is the fingerprint of a cationic intermediate: partial racemisation, rearranged skeletons, mixtures of S_N1 and E1 products, common-ion rate depression and σ^+ correlations. A rate law alone cannot prove a *free* cation: ion pairs and neighbouring groups often intervene, so the rate and the product-forming steps must be distinguished.

2.D1 tert-Butyl bromide in water: the full S_N1 sequence

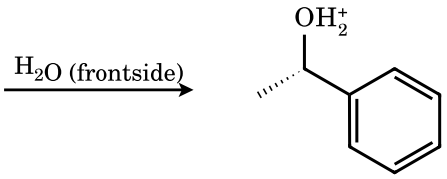


Rate = k[t-BuBr]. The cation is captured by the solvent (rate of capture ⩾ rate of formation, so capture is not kinetically visible) and a second water removes H^+. Competing E1 gives 2-methylpropene (≈ 10 % in water, more in ethanol or at higher temperature).

2.D2 Stereochemistry: (S)-1-phenylethyl chloride in aqueous acetone



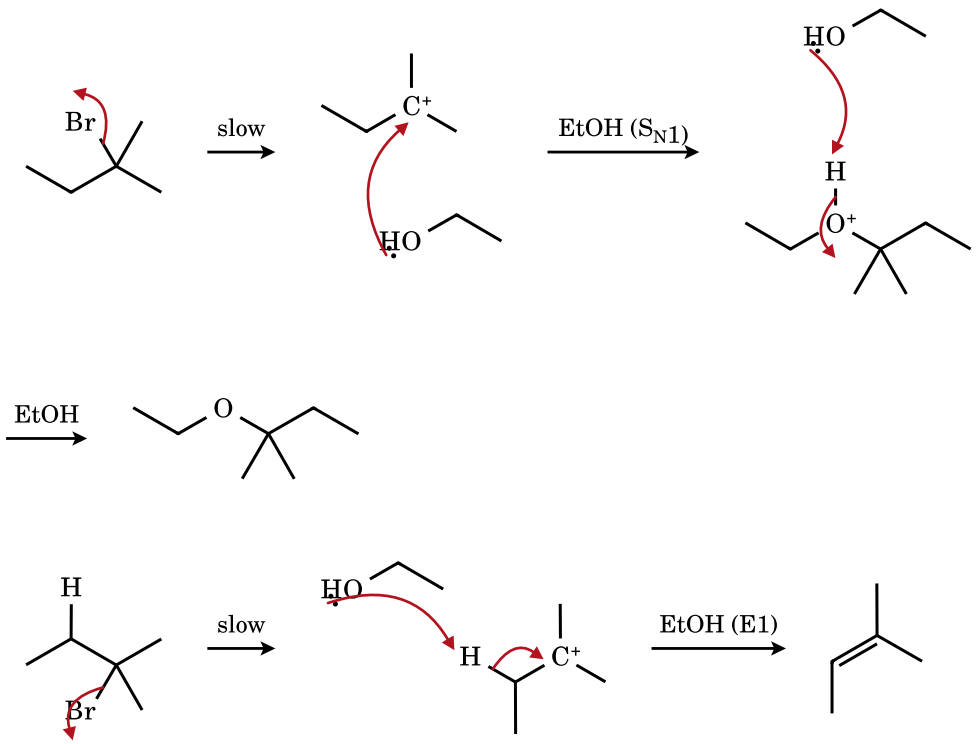
(S)-1-phenylethyl chloride **contact ion pair** **(R) inversion, major**



(S) retention, minor

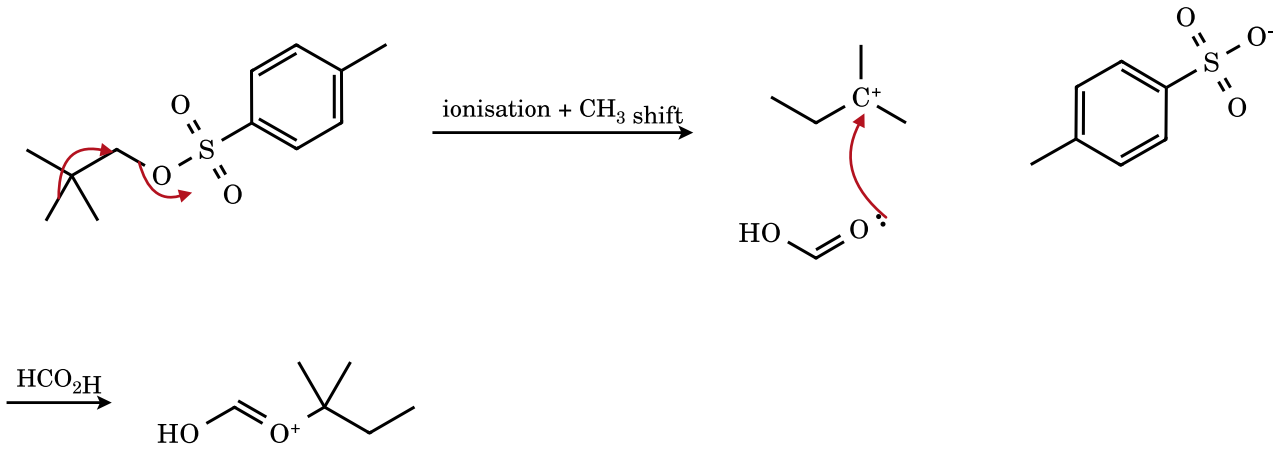
The planar cation can be attacked from either face, so racemisation is expected; observation gives *partial* racemisation with net inversion. The leaving chloride, still sitting on the front face in the contact ion pair, shields that face; the solvent attacks preferentially from the back. Free (solvent-separated) cations would give 50:50. The ratio is an experimental number, not a prediction.

2.D3 2-Bromo-2-methylbutane in ethanol: S_N1 and E1 from the same cation



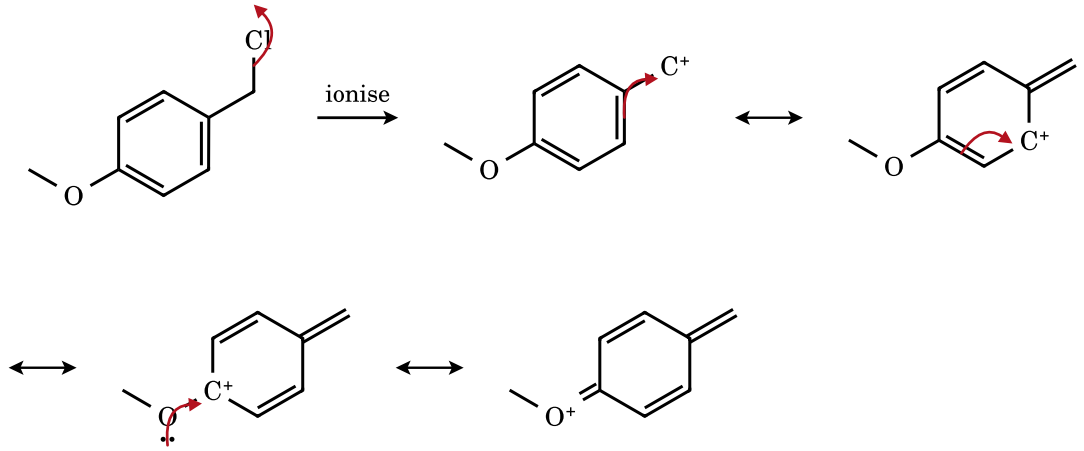
One cation, two destinations: nucleophilic capture at carbon (ether) or loss of a β-hydrogen (alkene). Higher temperature favours E1 (greater entropy gain). The more substituted alkene (2-methylbut-2-ene) dominates; the C–H bond that is lost must overlap the empty p orbital.

2.D4 Neopentyl tosylate in formic acid (formolysis): S_N1 with rearrangement



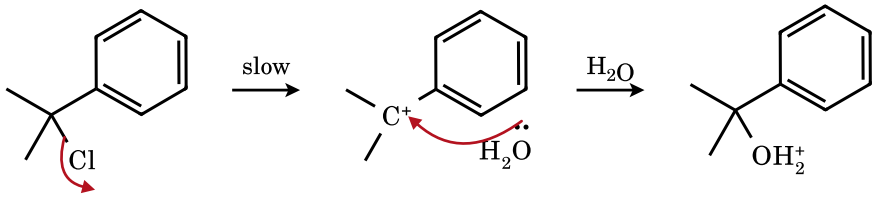
The only substitution product is tert-amyl formate (2-methylbut-2-yl formate); no neopentyl formate. A neopentyl system cannot do S_N2, so the ionising solvent pushes it through the σ-assisted pathway. The mixture also gives 2-methylbut-2-ene (E1).

2.D5 4-Methoxybenzyl chloride in water: why +M groups accelerate S_N1



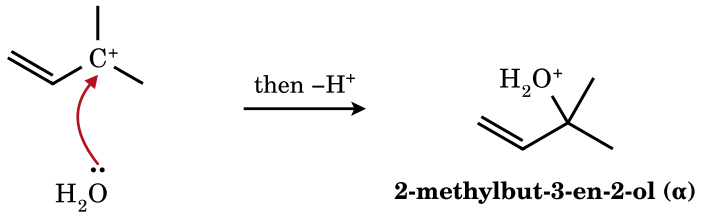
Ionisation puts + on the benzylic CH_2; three successive π-shifts carry it to C-4, where the OMe lone pair completes the octet: a p-quinone-methide oxocarbenium ion. The methoxy group lowers the barrier by tens of kJ mol^{-1} relative to H ($\sigma^+_{\text{p-OMe}} = -0.78$).

2.D6 2-Chloro-2-phenylpropane (cumyl chloride) in 90 % aqueous acetone



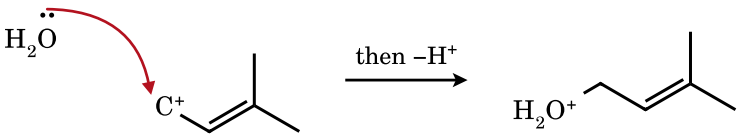
The tertiary benzylic cation is the standard system used to define σ^+ substituent constants (Brown–Okamoto, $\rho \approx -4.5$): the very large negative ρ shows that substantial positive charge is delocalised into the ring in the transition state.

2.D7 Prenyl chloride in water: S_N1 and S_N1' products from one allyl cation



2-methylbut-3-en-2-ol (α)

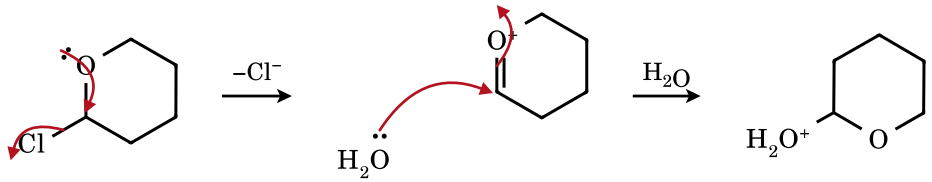
attack at C-2 (tertiary)



attack at C-4 (primary) **prenol (γ)**

The delocalised cation is attacked at either terminus, and both isomeric chlorides give the same two alcohols. Under kinetic control the tertiary alcohol is the major product (most of the positive charge resides on the tertiary carbon); prenol, which has the more substituted alkene, accumulates only if the tertiary alcohol can re-ionise (acidic, longer reaction times) – kinetic versus thermodynamic control (Part 9).

2.D8 2-Chlorotetrahydropyran: α-oxygen stabilisation

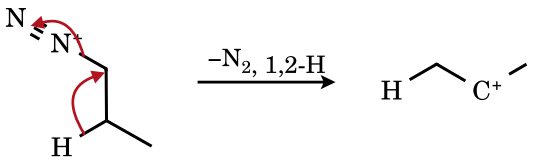


Glycosyl-type α-halo ethers ionise with ring-oxygen assistance (the axial lone pair overlaps $\sigma^*_{C–Cl}$, the anomeric effect in action). The cyclic oxocarbenium ion is attacked from either face: this is the basis of glycoside synthesis (Koenigs–Knorr, with Ag^+ activation).

2.E Other routes: diazonium ions, Lewis acids, hydride abstraction, acid-mediated ionisations

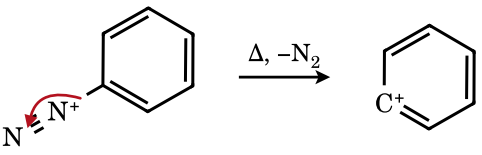
Some cations are produced neither from an alcohol/halide nor from an alkene. The common feature is a very good leaving group (N_2, H_2, H_2O, ROH) or a very strong electrophile (H^+ in superacid, Ph_3C^+, $AlCl_3$). Radical and concerted pathways are *not* counted here: e.g. the Hofmann–Löffler or Kolbe reactions are radical, and Tiffeneau–Demjanov, Curtius and Hofmann reactions proceed with migration concerted with loss of N_2 (no free cation).

2.E1 n-Propylamine + HNO_2 (via n-propyldiazonium)



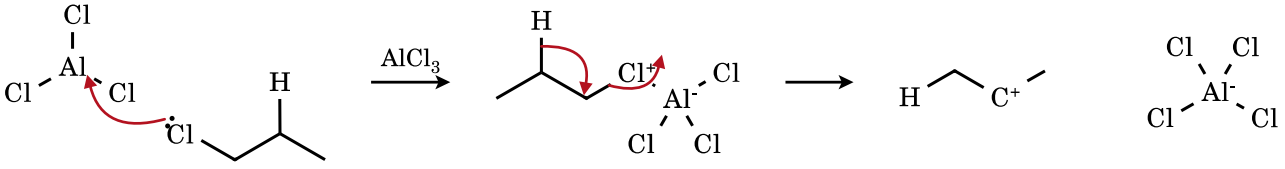
Alkyldiazonium ions are the most reactive leaving-group systems known (N_2, a thermodynamically very stable molecule). Even a primary one gives a mixture in which the 2-propyl derivatives (2-propanol, 2-chloropropane) outweigh the 1-propyl ones, together with propene, because the departing N_2 is pushed out by a neighbouring migrating H or alkyl group. The result is a classic reason why nitrous acid deamination of primary amines is synthetically useless.

2.E2 Benzenediazonium ion in hot water: aryl cation



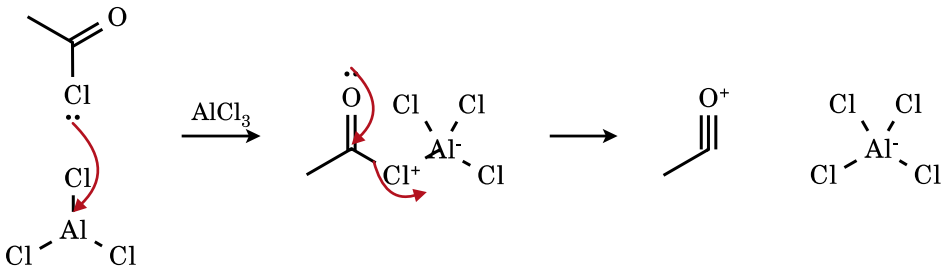
The aryl cation has its empty orbital in the plane of the ring (an sp^2 σ orbital orthogonal to the π system), so it gets no resonance help and is among the least stable carbocations; it is formed only because N_2 is such a good leaving group. It reacts with the nearest nucleophile (H_2O → phenol), even F^- in the Balz–Schiemann reaction.

2.E3 n-Propyl chloride + $AlCl_3$: 1,2-hydride shift during ionisation



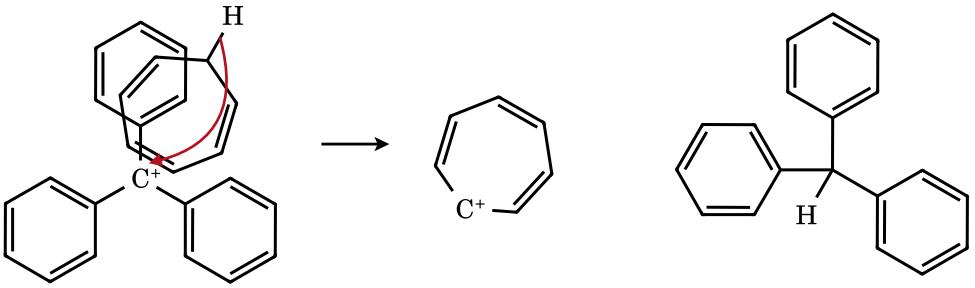
The polarised complex R–Cl→$AlCl_3$ transfers a hydride to the carbon that is losing Cl^-: isopropyl cation (as an ion pair with $AlCl_4^-$). This is why benzene + n-propyl chloride/$AlCl_3$ gives cumene, not n-propylbenzene.

2.E4 Acetyl chloride + $AlCl_3$: acylium ion



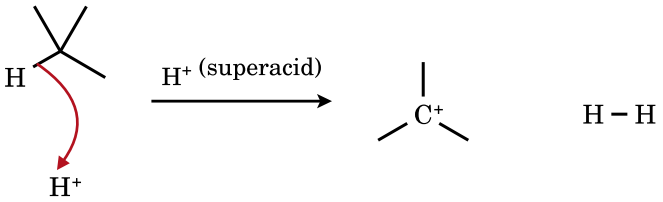
The oxygen lone pair drives out chloride and forms a third C–O bond; every atom in $CH_3–C \equiv O^+$ has an octet. Acylium ions are linear (sp-hybridised C) and do **not** rearrange, so Friedel–Crafts acylation gives unrearranged ketones; the ketone product deactivates the ring, which is why acylation (unlike alkylation) stops at one substitution.

2.E5 Hydride abstraction: cycloheptatriene + $Ph_3C^+\ BF_4^-$



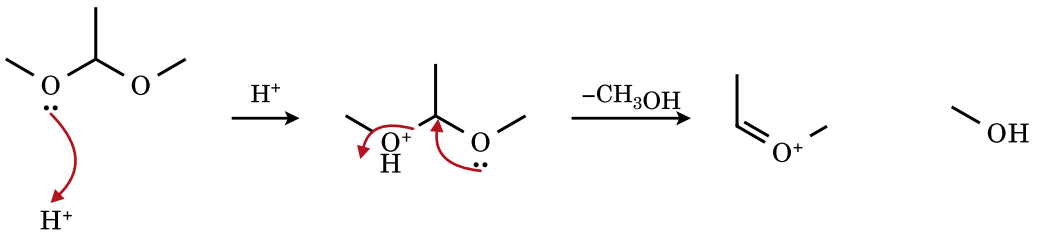
The C–H σ pair of the sp^3 CH_2 transfers as H^- to the trityl cation (an equilibrium that lies far to the right because the product is the aromatic 6π tropylium ion, $pK_{R+} \approx +4.7$, against −6.6 for trityl). The seven-membered ring is planar, all seven C–C bonds are equal, and the positive charge is shared equally.

2.E6 Isobutane in $FSO_3H{\cdot}SbF_5$ (magic acid): σ-protonation then loss of H_2



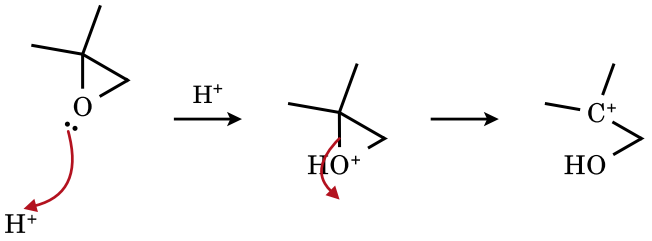
A C–H σ bond is a very weak nucleophile, but the proton of a superacid attacks it, forming a three-centre two-electron bond (a *carbonium* ion, $R_3C(H_2)^+$, see 1.4). Loss of H_2 leaves the carbenium ion. This is the basis of H/D exchange of alkanes in superacid and of alkane isomerisation (Olah).

2.E7 Acid hydrolysis of acetaldehyde dimethyl acetal



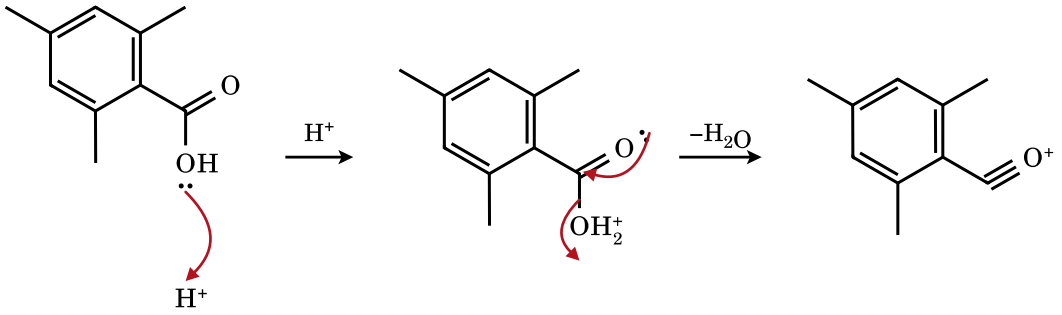
Protonation of one OMe turns it into a good leaving group (CH_3OH); the lone pair of the other oxygen pushes it out. The oxocarbenium ion is the intermediate that water attacks in the rate-limiting or following step: acetals are stable to base but labile to dilute acid.

2.E8 Acid-catalysed opening of 2,2-dimethyloxirane



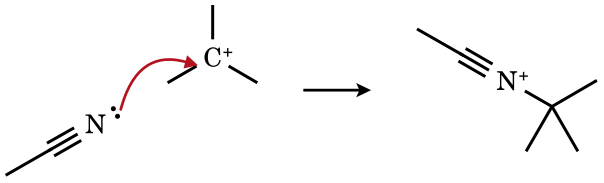
Protonation makes the oxirane a good electrophile; the C–O bond to the *more substituted* carbon is longer and weaker and breaks first, so the nucleophile (H_2O, ROH, X^-) ends up on the tertiary carbon (contrast base-promoted opening, S_N2 at the less hindered CH_2). The true transition state is a loose S_N2 with S_N1 character ('borderline'); a full tertiary cation is shown to indicate the charge distribution.

2.E9 2,4,6-Trimethylbenzoic acid in conc. H_2SO_4: $A_{AC}1$ acylium



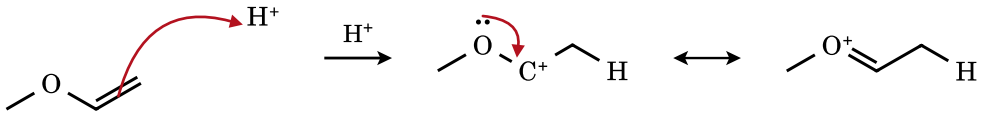
Ordinarily esterification/hydrolysis of benzoic acids goes by addition–elimination ($A_{AC}2$) through a tetrahedral intermediate. The two ortho methyl groups block nucleophilic addition to C=O, so the reaction switches to ionisation to a mesitoyl acylium ion, which is linear and so relieves the steric clash. Dissolving in H_2SO_4 and pouring into alcohol gives the ester in high yield (Newman).

2.E10 Ritter reaction: nitrile + tert-butyl cation



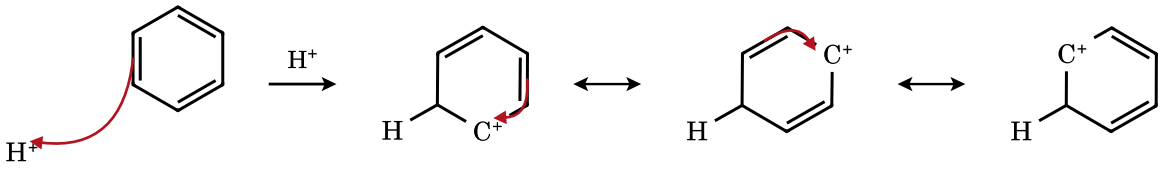
A nitrile nitrogen lone pair (sp-hybridised, weakly basic but unhindered) captures the cation: the nitrilium ion R–C ≡ N^+–R′ is attacked by water at carbon to give an N-tert-butylamide. Only cations that form readily (tertiary, benzylic) work.

2.E11 Methyl vinyl ether + H⁺: protonation of an enol ether



Protonation of the β-carbon gives a cation that is immediately an oxocarbenium ion because the α-oxygen lone pair donates into the empty p orbital. Enol ethers are hydrolysed orders of magnitude faster than simple alkenes are hydrated for this reason (the rate-limiting step is protonation at carbon).

2.E12 Benzene + H⁺ (superacid): the arenium (Wheland) ion



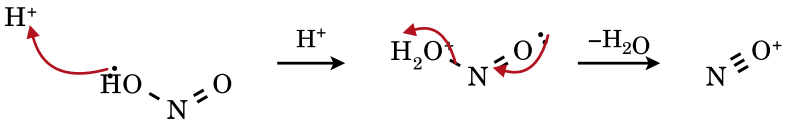
$C_6H_7^+$ is the intermediate in every electrophilic aromatic substitution. The positive charge is delocalised over the ortho and para carbons only (three contributors; the meta carbons carry none); the sp^3 CH_2 breaks the aromatic ring, which is why this step costs so much energy and is rate-determining.

2.F Carbocations from amines: diazotisation with nitrous acid

Nitrous acid (from $NaNO_2$ + HX, never stored) is the reagent that turns a **primary** amino group ($–NH_2$), the worst of leaving groups, into the best known one (N_2). The active electrophile in dilute mineral acid is the **nitrosonium ion NO^+**. What happens next depends on the type of amine:

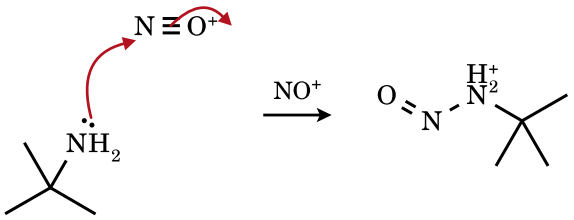
Amine	$NaNO_2$ / HCl, 0–5 °C	Carbocation?
Primary aliphatic RCH_2NH_2, R_3CNH_2	$R–N_2^+$ (not isolable) → loses N_2 at once	**Yes** (or a concerted shift/substitution with N_2 loss); mixture of alcohols, alkenes, halides; rearranged products common
Primary aromatic $ArNH_2$	$Ar–N_2^+$ $X^–$: stable in cold solution (resonance with the ring)	**No** at 0–5 °C. On warming in water the diazonium ion loses N_2 to give the aryl cation (phenol)
Primary benzylic $ArCH_2NH_2$	$ArCH_2N_2^+$ (not stable)	**Yes**: benzylic cation $ArCH_2^+$
Secondary R_2NH (and ArNHR)	N-nitrosamine $R_2N–N=O$ (yellow oil)	**No** (no N–H left to lose from N after nitrosation; the N–N bond is not an N_2-releasing diazonium)
Tertiary R_3N	R_3NH^+ $NO_2^–$ (salt); no stable product	**No** (tertiary aryl amines such as $PhNMe_2$ undergo C-nitrosation at the para carbon)

2.F1 Nitrosonium ion from nitrous acid



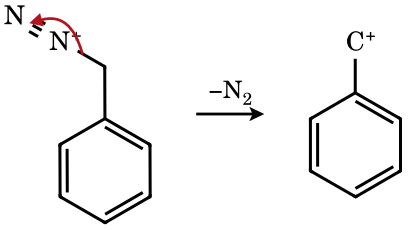
Protonation of the hydroxyl oxygen turns it into water; the oxygen lone pair of the N=O then forms a third N–O bond as water leaves, giving $N \equiv O^+$ (isoelectronic with CO and N_2). NO^+ is the electrophile that attacks amines (N-nitrosation).

2.F2 tert-Butylamine + NO^+: N-nitrosation (first step)



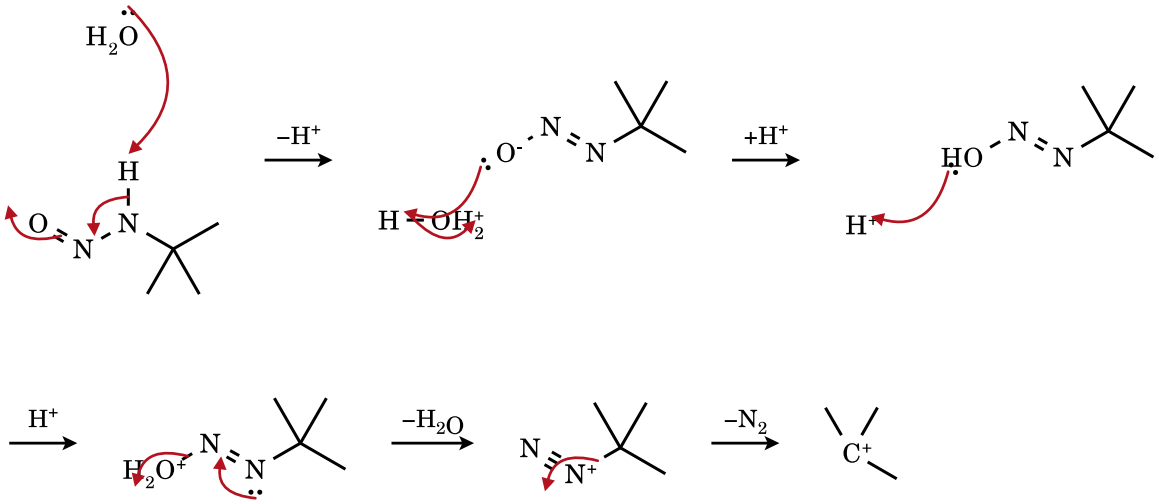
The amine nitrogen lone pair attacks nitrogen of NO^+; one N ≡ O π bond moves onto oxygen. The N-nitrosoammonium ion then loses H^+ and tautomerises, as drawn in 2.F3.

2.F4 Benzylamine + HNO_2: benzylic cation



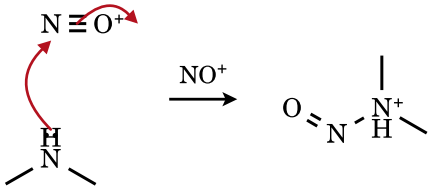
Benzyldiazonium ions lose N_2 at once, even at 0 °C, because the benzylic cation is resonance-stabilised. The cation is captured by water (benzyl alcohol); compare ArN_2^+, where the diazonium group is attached to an sp^2 carbon and is stable in the cold.

2.F3 …from N-nitroso-tert-butylammonium to the cation (R–NH–N=O → R–N=N–OH → R–N_2^+ → R^+)



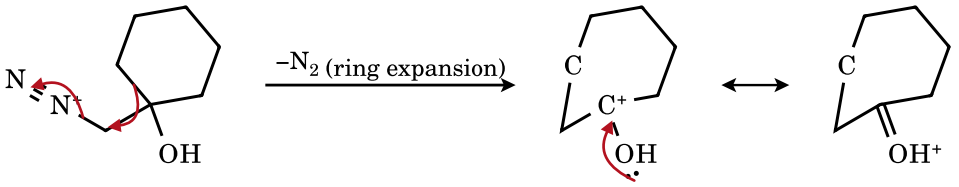
Loss of N–H proton and shift of the π bond gives the diazotate; protonation on oxygen (diazohydroxide R–N=N–OH), then a second protonation and loss of water gives the alkyldiazonium ion R–N ≡ N^+; its C–N σ pair leaves with N_2. For a tertiary alkyl group the product is the tert-butyl cation (captured by water or losing H^+). With a primary alkyl group the same sequence is followed by a concerted shift (2.E1).

2.F5 Secondary amine + NO^+: N-nitrosamine (no carbocation)



The nitrosoammonium ion can only lose its N–H proton to give R_2N–N=O; there is no hydrogen to move to oxygen to form a diazonium, so the sequence stops at the nitrosamine (a carcinogen class). No carbocation forms.

2.F6 Tiffeneau–Demjanov: 1-(aminomethyl)cyclohexanol + HNO_2 (diazonium, then ring expansion)

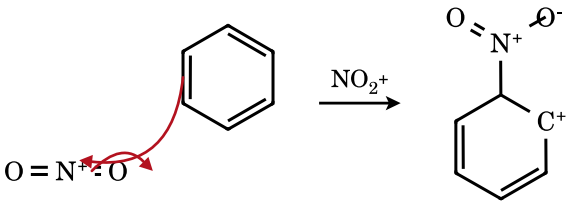


The ring C–C bond anti to the departing N_2 migrates to the CH_2 as N_2 leaves, so the six-membered ring becomes seven-membered; the cation produced is the protonated ketone (oxygen lone pair donates), which loses H^+ to give cycloheptanone. A free primary cation is never formed.

2.G Carbocations from aromatic compounds: arenium (σ-complex, Wheland) ions

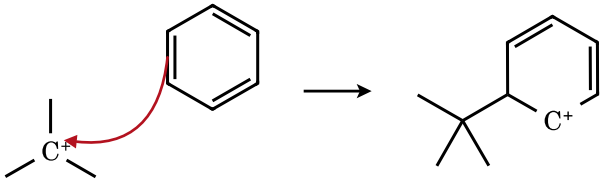
An aromatic ring is a poor nucleophile because attack on an electrophile E^+ destroys aromaticity. The product is a **cyclohexadienyl (arenium) cation**: one sp^3 carbon carrying E and H, and the other five carbons conjugated over **three contributors** with the charge at the two *ortho* and the *para* positions relative to the sp^3 carbon (never the meta carbons). Loss of H^+ from the sp^3 carbon restores the aromatic ring (substitution). Formation of the arenium ion is the slow step in every electrophilic aromatic substitution.

2.G1 Nitration: benzene + NO_2^+ (HNO_3/H_2SO_4)



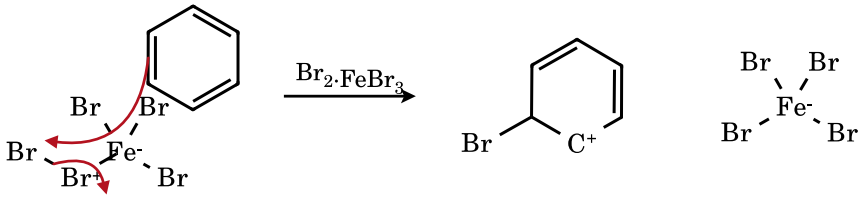
The nitronium ion (O=N⁺=O, linear) is generated by H_2SO_4 protonating HNO_3 (HNO_3 + 2 H_2SO_4 → NO_2^+ + H_3O^+ + 2 HSO_4^-). Its attack on benzene gives the nitro-substituted arenium ion; HSO_4^- removes the ring proton.

2.G3 Friedel–Crafts alkylation: benzene + $t\text{-}Bu^+$ ($t\text{-}BuCl/AlCl_3$)



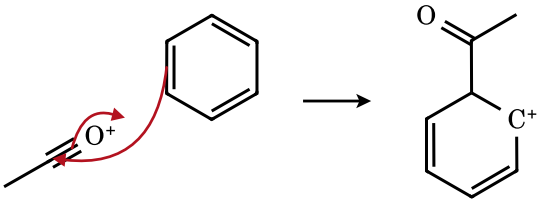
A carbocation (from R–Cl + $AlCl_3$, from an alkene + H^+, or from an alcohol + acid) is attacked by the arene. Cations that can rearrange (n-propyl, n-butyl, neopentyl…) give rearranged alkylbenzenes: cumene, not n-propylbenzene, from n-PrCl/$AlCl_3$. The product ring is more reactive than benzene, so polyalkylation is a problem.

2.G2 Bromination: benzene + $Br_2/FeBr_3$



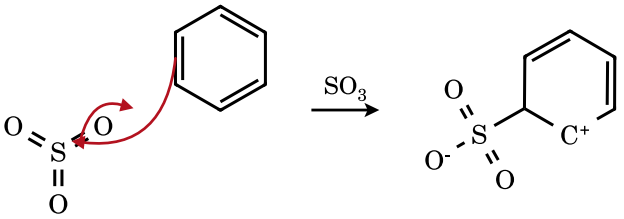
$FeBr_3$ polarises Br–Br ($Br^{\delta+}\cdots Br\text{–}FeBr_3^{\delta-}$); the π electrons attack the electrophilic bromine and displace $[FeBr_4]^-$. The same arenium ion forms regardless of which Br leaves with the Lewis acid.

2.G4 Friedel–Crafts acylation: benzene + CH_3CO^+ (acylium ion)



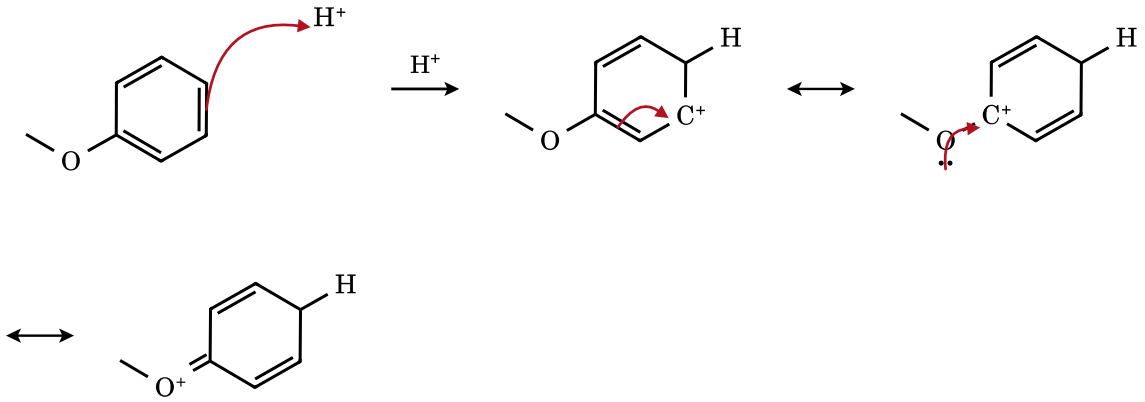
The acylium ion adds to the ring through its electrophilic carbon (one C ≡ O π bond becomes the C=O of the ketone). The acyl group deactivates the ring, so one substitution only, and acylium ions do not rearrange.

2.G5 Sulfonation: benzene + SO_3 (oleum)



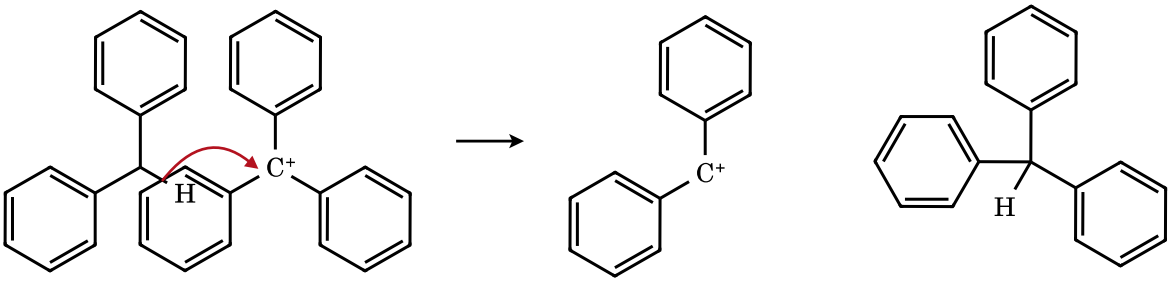
SO_3 (a neutral, very electrophilic sulfur) is attacked by the π bond; the S=O π electrons move onto oxygen, giving a zwitterionic arenium ion. Sulfonation is reversible (dilute hot acid removes SO_3H), unlike nitration or halogenation.

2.G6 Protonation of anisole at the para position (ipso to H)



Protonating C-1 (para to OMe) puts positive charge at the ortho and para carbons, and one contributor places it on C-4, which carries OMe: the oxygen lone pair then delocalises it (an oxocarbenium contributor). Protonation meta to OMe lacks this contributor, so ortho/para attack is faster – the origin of ortho/para direction by +M groups.

2.G7 Hydride abstraction from an alkylarene: diphenylmethane + Ph_3C^+



Aryl groups on a C–H carbon make it a hydride donor: loss of H^- leaves the diarylmethyl (benzhydryl) cation, stabilised by two rings. Ph_3C^+ is a hydride acceptor only for strongly donating hydrocarbons (cycloheptatriene, 2.E5); with Ph_2CH_2 the equilibrium lies far on the left (pK_{R+} –6.6 versus –13.3).

Part 3 Leaving-Group Ability

3.1 What a leaving group is, and what 'good' means

A **leaving group** (nucleofuge) is the fragment that departs with the bonding electron pair in a heterolytic substitution or elimination: R–LG → R^+ + LG^- (or neutral LG from a cationic substrate). 'Good' is a statement about a *rate* (kinetic, a lowered barrier for the C–LG cleavage step) that correlates with – but is not identical to – the *thermodynamic* stability of the departed LG, i.e. the acidity of its conjugate acid H–LG. The correlation exists because in the transition state the LG carries a partial negative charge; the better it can stabilise that charge, the lower the barrier (Brønsted-type relationship, typically β_{lg} between −0.3 and −1.0 for substitution).

Leaving group (LG)	Conjugate acid	pK_a (aq., approx.)	Remarks
I^-, Br^-, Cl^-	HI, HBr, HCl	−10, −9, −7	large, polarisable; weak C–X bonds (I < Br < Cl)
F^-	HF	3.2	very poor in S_N1/S_N2 (strong C–F bond); good in SNAr (see Q5)
$CF_3SO_3^-$ (triflate)	CF_3SO_3H	≈ −14	charge delocalised over 3 O, CF_3 –I effect; a 'super' LG
p-$O_2NC_6H_4SO_3^-$ (nosylate)	$ArSO_3H$	≈ −6.5 (est.)	EWG on the aryl ring improves LG
p-$CH_3C_6H_4SO_3^-$ (tosylate)	TsOH	≈ −2.8	standard crystalline sulfonate
$CH_3SO_3^-$ (mesylate)	MsOH	≈ −1.9	similar to tosylate in rate
H_2O (from ROH_2^+)	H_3O^+	−1.7	neutral LG; HO^- of ROH is not a leaving group
$CF_3CO_2^-$, $CH_3CO_2^-$	CF_3CO_2H, AcOH	0.5, 4.76	carboxylates: moderate; acetates ionise only for cations of tertiary/benzylic stability
N_2 (from RN_2^+)	—	—	best known LG: very stable neutral molecule, huge entropy gain
NH_3 (from RNH_3^+)	NH_4^+	9.25	poor: Hofmann elimination needs heat; NH_2^- (pK_a 38) never leaves
HO^-, RO^-, H^-, R^-	H_2O, ROH, H_2, RH	15.7, 16–19, 35, ≈ 50	do not leave in S_N/E1 unless activated (E1cB, intramolecular, ring strain)

3.2 Factors that determine leaving ability

- **Basicity / conjugate-acid strength.** Weaker bases are better leaving groups: the LG^- is more stable. Within a period this is the dominant trend (I^- ⋙ F^- in protic solvent; TfO^- ⋙ AcO^- ⋙ HO^-).
- **Bond strength and polarisability.** Down a group the C–X bond weakens (C–F ≈ 110, C–Cl ≈ 84, C–Br ≈ 70, C–I ≈ 57 kcal mol^{-1}) and the large, soft anion is stabilised by polarisation of the TS; the pK_a trend (HF 3.2 → HI −10) reinforces this. In the gas phase F^- is a worse LG than I^- for the same reason.
- **Resonance and charge delocalisation.** Sulfonates (charge over 3 O), carboxylates (2 O), triflate (3 O + CF_3) are good LGs because the charge is spread; alkoxides and amides are not.

- **Induction.** EWGs on the LG (CF_3, NO_2, halogens on an aryl ring) stabilise the anion: $TfO^- \gg MsO^-$; p-O_2N–$C_6H_4SO_3^-$ > TsO^-.
- **Solvation.** Protic solvents hydrogen-bond to small, charge-dense anions (F^-, Cl^-, RO^-), stabilising them in the TS and promoting ionisation; the effect on the *ordering* of halides is small for S_N1 but large for nucleophiles in S_N2. Aprotic dipolar solvents (DMSO, DMF) favour S_N2 by leaving anions bare.
- **Neutral versus charged substrates.** Protonation, alkylation or diazotisation converts a bad LG (HO^-, NH_2^-) into a good neutral one (H_2O, NH_3, N_2) – the underlying reason for acid catalysis and for conversion of alcohols to sulfonates.
- **Neighbouring-group assistance.** A lone pair or π/σ bond anti-periplanar to the C–LG bond can displace the LG intramolecularly (anchimeric assistance; sulfur > nitrogen-free base > bromine > oxygen in typical hydrolyses): the C–LG bond breaks faster than the intrinsic LG ability would suggest.
- **Stereoelectronic and steric factors.** An axial LG on a cyclohexane ring is anti-periplanar to ring C–C/C–H bonds and relieves 1,3-diaxial strain on ionisation; an anti-periplanar C–Si bond (β-silicon effect) accelerates departure by orders of magnitude; large groups on the substrate relieve strain (B-strain) on ionisation.
- **Ion-pair behaviour.** The LG may remain close to the cation (internal return). Poor ion-pair return from large, delocalised LGs (OTs, OTf) increases the net rate relative to Cl^-.

3.3 Why there is no single universal order

> **The order I^- > Br^- > Cl^- ≫ F^- holds for S_N1, S_N2 and E2 at sp^3 carbon, but is reversed (F ≫ Cl ≈ Br > I) in nucleophilic aromatic substitution**, where C–LG cleavage is *not* rate-limiting: the first step (attack of the nucleophile to give the Meisenheimer complex) is slower, and the very electronegative fluorine stabilises that TS by induction. Likewise in acyl substitution the order of acyl derivatives (RCOCl > $(RCO)_2O$ > RCO_2R' > $RCONR_2$) reflects both LG ability *and* how strongly the LG donates into C=O. In E2 the LG changes the *regiochemistry*: poorer LGs (F^-, NR_3) make the TS carbanion-like and favour the Hofmann (less substituted) alkene. And in transition-metal oxidative addition the order is Ar–I > Ar–Br ≈ Ar–OTf > Ar–Cl, controlled by bond strength and metal–LG interaction, not by pK_a.

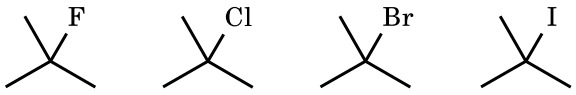
The following comparisons are made at stated conditions. A leaving-group ranking is meaningful only for a **fixed substrate, mechanism, solvent and temperature**.

Process	Rate-limiting step	Leaving-group dependence
S_N1 / E1	C–LG heterolysis	Strong (β_{lg} ≈ −0.5 to −1); solvent H-bond donation important
S_N2	concerted attack + C–LG cleavage	Moderate (β_{lg} ≈ −0.3 to −0.6); solvent-nucleophile interactions also matter
E2	concerted; anti-periplanar H–C–C–LG	Moderate; also changes Zaitsev/Hofmann ratio and TS looseness
E1cB	carbanion formation or LG expulsion	LG may be as poor as HO^- or RO^- if carbanion is stabilised (enolate)
SNAr	nucleophilic addition	Weak, ‘element effect’ F ≫ Cl ≈ Br > I
Acyl substitution	addition or collapse of tetrahedral intermediate	Strong, but coupled with resonance donation

3.4 Fifteen leaving-group comparison problems

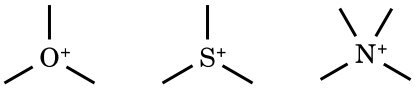
Rank **A–D** as instructed. Every statement of order refers to the stated conditions only. Rate orders are established qualitative trends in the literature; no numerical ratios are claimed unless stated.

LG1. Arrange A–D in **decreasing** first-order solvolysis rate in 80 % aq. ethanol at 25 °C (same alkyl group; the only variable is the halogen).



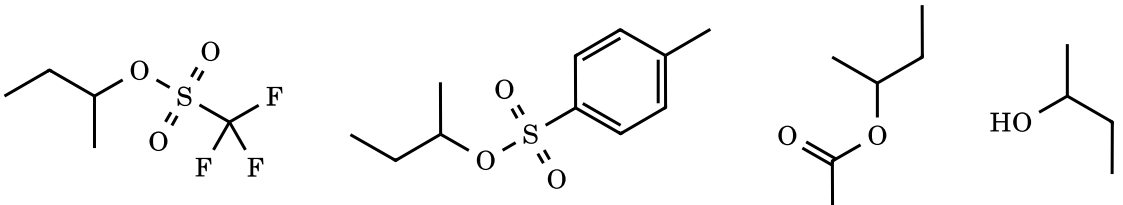
A **B** **C** **D**

LG10. Methyl-transfer (S_N2) from the onium salts A–C to a thiolate in CH_3CN at 25 °C. Arrange in **decreasing** reactivity.



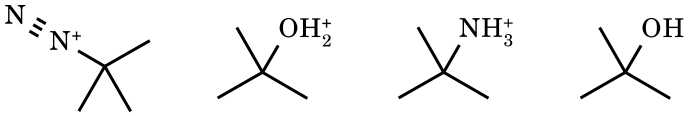
A **B** **C**

LG2. R = 2-butyl in all four. Arrange the derivatives in **decreasing** rate of ionisation in warm (50 °C) neutral ethanol (no acid added).



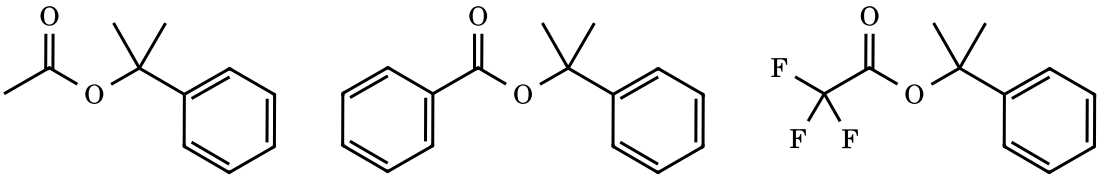
A **B** **C** **D**

LG3. For the tert-butyl derivatives A–D in acidic aqueous solution at 25 °C, arrange in **decreasing** ease of C–LG heterolysis (LG = the neutral molecule that leaves, or HO^- for D).

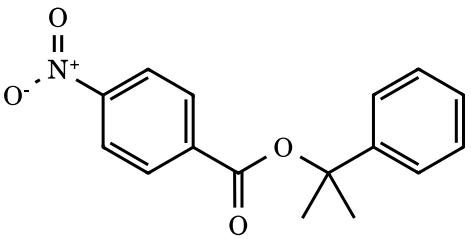


A **B** **C** **D**

LG4. The four tert-cumyl esters A–D are solvolysed in 70 % aq. acetone at 25 °C. Arrange in **decreasing** rate of ionisation.

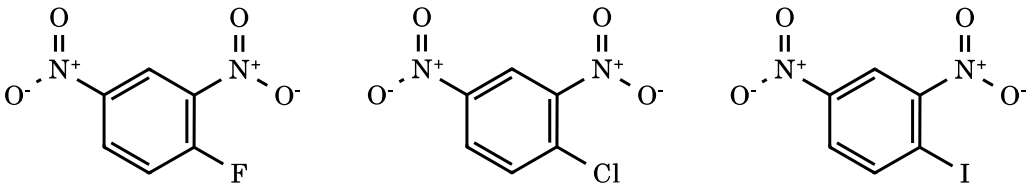


A **B** **C**



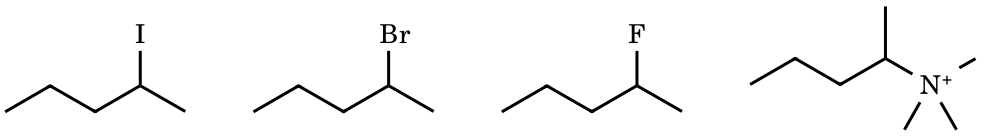
D

LG5. Arrange the 1-halo-2,4-dinitrobenzenes A–C in **decreasing** rate of reaction with piperidine (or methoxide) in ethanol at 25 °C.



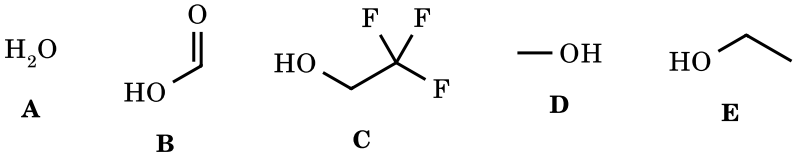
A **B** **C**

LG6. Each 2-pentyl derivative A–D is treated with NaOEt/EtOH (E2). Arrange in **decreasing** percentage of the *less substituted* alkene, pent-1-ene (Hofmann product), in the product mixture.



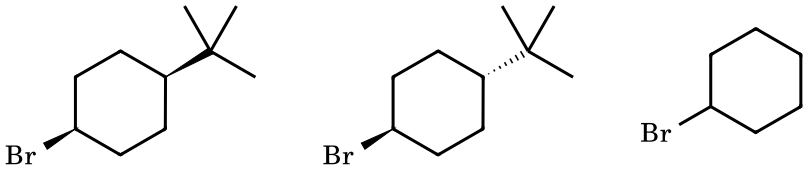
A **B** **C** **D**

LG7. Arrange the solvents A–E in **decreasing** ionising power for the solvolysis of tert-butyl chloride at 25 °C (Grunwald–Winstein Y scale). Solvents: A H_2O, B HCO_2H, C CF_3CH_2OH, D CH_3OH, E CH_3CH_2OH.



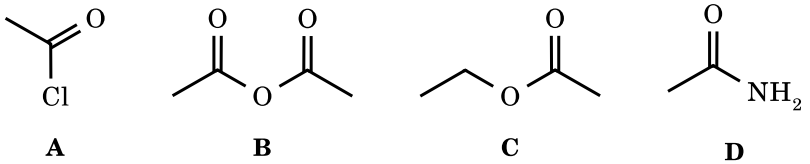
A **B** **C** **D** **E**

LG8. Bromides A–C undergo E2 with NaOEt in EtOH at 50 °C. Arrange in **decreasing** rate. (A cis-4-tert-butylcyclohexyl bromide; B trans-4-tert-butylcyclohexyl bromide; C bromocyclohexane. The t-Bu group is locked equatorial.)



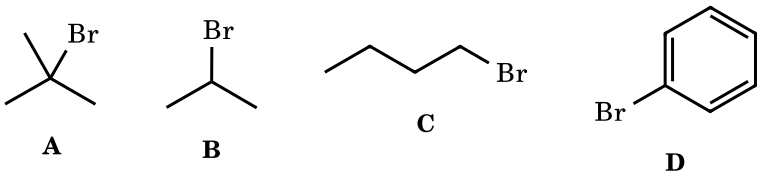
A cis **B trans** **C**

LG9. Arrange the acetyl derivatives A–D in **decreasing** rate of nucleophilic acyl substitution by water (neutral, 25 °C).



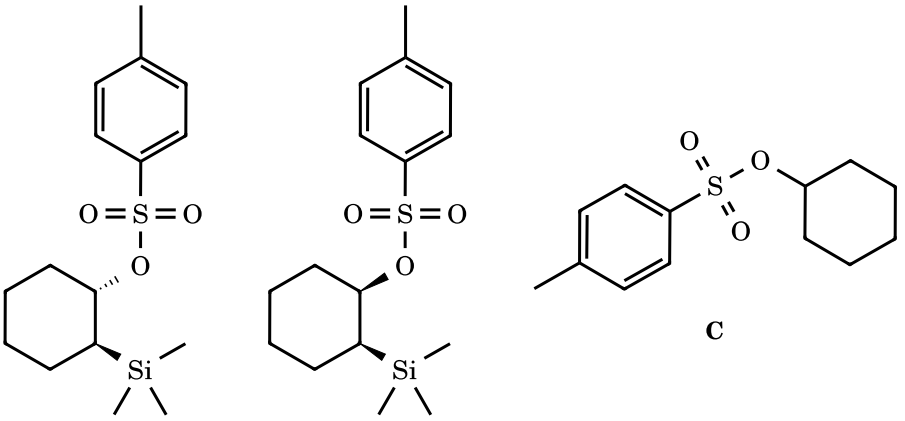
A **B** **C** **D**

LG11. Treat each compound A–D with ethanolic $AgNO_3$ at 25 °C. Arrange in **decreasing** rate of AgBr precipitation (no precipitate = last).



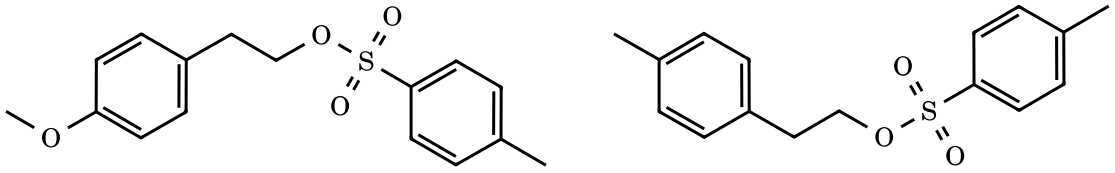
A **B** **C** **D**

LG12. Tosylates A–C are acetolysed at 50 °C. Arrange in **decreasing** rate. (A trans-2-(trimethylsilyl)cyclohexyl tosylate; B cis-2-(trimethylsilyl)cyclohexyl tosylate; C cyclohexyl tosylate.)



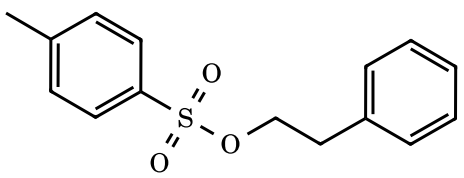
A trans **B cis** **C**

LG13. Arrange the 2-arylethyl tosylates A–D in **decreasing** rate of acetolysis at 75 °C (neighbouring-group participation by the aryl ring is the major pathway).

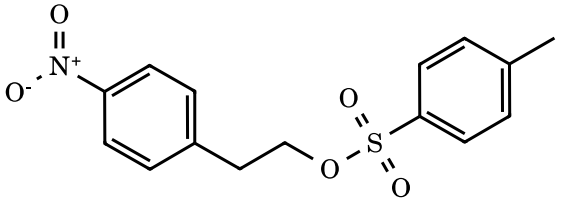


A

B

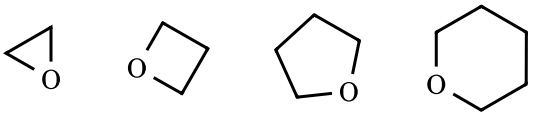


C



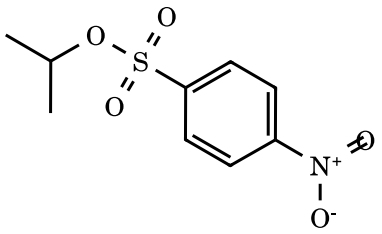
D

LG14. Arrange the cyclic ethers A–D in **decreasing** ease of acid-catalysed ring opening by aqueous HBr at 25 °C.

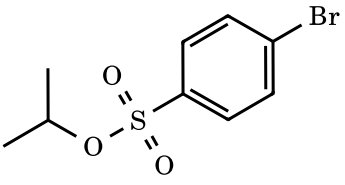


A **B** **C** **D**

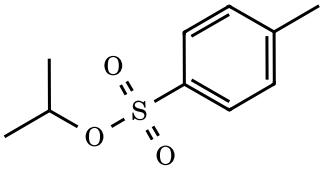
LG15. Arrange the 2-propyl arenesulfonates A–D in **decreasing** rate of solvolysis in aq. ethanol at 25 °C.



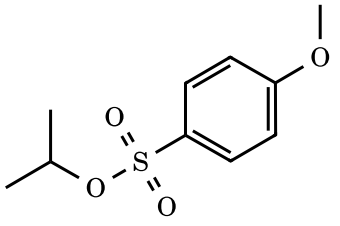
A p-NO_2



B p-Br



C p-CH_3



D p-OCH_3

Part 4 Carbocation Stability: Complete Advanced Treatment

What 'stability' means. Four different observables are routinely mixed up and must be kept apart: (1) *gas-phase thermodynamic stability*, measured by the hydride ion affinity (HIA) $R^+ + H^- \rightarrow R–H$ (smaller HIA = more stable cation) or the corresponding calculated energies – this reflects the *intrinsic electronic* stabilisation, free of solvent; (2) *stability in solution toward water*, pK_{R+} = pH at which $[R^+] = [ROH]$, applicable only to very stable cations; (3) *substituent stabilisation* expressed by σ^+ constants (Brown–Okamoto), derived from the solvolysis of $ArC(CH_3)_2Cl$; (4) *ionisation rates* (solvolysis), which depend on the stability of the cation *and* on the leaving group, solvent, ion-pair return, nucleophilic solvent assistance (k_s), neighbouring-group participation (k_Δ) and strain relief, and therefore track the cation's stability only when all of these are held constant and the transition state is late. Every problem in Part 5 states which observable is meant.

> Orders obtained in the gas phase can differ from, or even invert, orders in solution (alkyl-substituted benzyl cations: gas phase t-Bu > i-Pr > Et > Me, solution Me ≳ Et > i-Pr > t-Bu – the Baker–Nathan order – because larger alkyl groups are solvated less effectively). Where an order below depends on the medium, the medium is stated.

4.1 Inductive effect

A σ-framework effect: electronegativity differences polarise σ bonds. **+I** (electron-donating) groups – alkyl groups, and especially more polarisable and more electropositive groups such as $SiMe_3$, $SnMe_3$ – push electron density toward the cationic carbon and lower its energy; **–I** groups (F, Cl, OR, NR_3^+, NO_2, CN, CF_3, C=O) pull density away and raise it. The effect falls off roughly by a factor of 2–3 per intervening bond, so it is dominant for α and β substituents, small for γ. The field (through-space) component is important in rigid frameworks (4-substituted bicyclo[2.2.2]octyl systems). **Important caveat:** –I groups attached by a lone-pair atom (OR, SR, NR_2, Hal) at the α position are overall *stabilising*

because the +M effect outweighs –I ($CH_3OCH_2^+$ is far more stable than $CH_3CH_2^+$), while the same groups at β position are destabilising unless they can bridge. Strongly –I groups on the cationic carbon are dramatically destabilising: α-CF_3 and α-carbonyl groups reduce solvolysis rates of tertiary systems by many orders of magnitude (Tidwell).

4.2 Mesomeric (resonance) effects

A π-bond or lone pair attached to the cationic carbon delocalises the charge (π→p or n→p overlap). **+M** groups (NR_2 > NHR > OH ≈ OR > SR > aryl ≈ vinyl > halogen) stabilise; **–M** groups (C=O, CN, NO_2, SO_2R, CF_3 by negative hyperconjugation/induction) destabilise by competing for the same orbital or by withdrawing π density. Resonance changes *only electron positions* and is shown with the double-headed arrow ⇔; curved arrows show how to go from one contributor to the next.

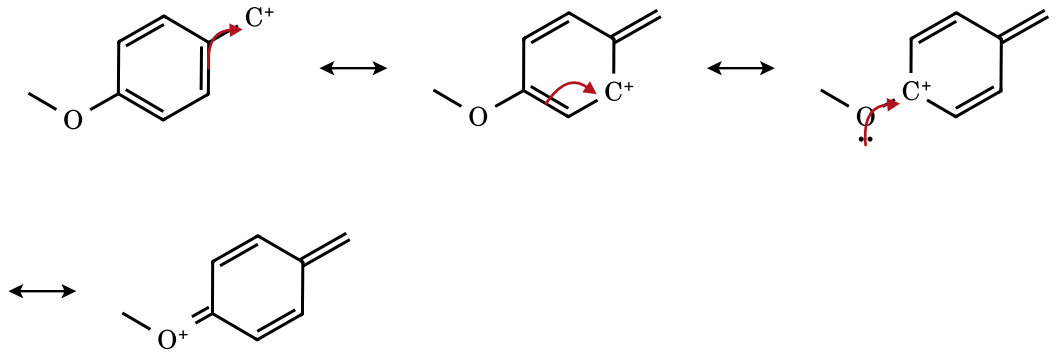


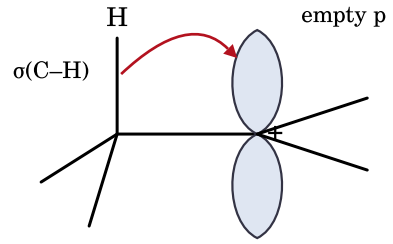
Figure 4.1 p-Methoxybenzyl cation: charge delocalised from CH_2 to the ortho/para carbons, and finally onto oxygen (an oxocarbenium contributor; every atom has an octet).

- **Position matters.** Charge in a benzyl cation sits on the ortho and para carbons, not meta. A +M group at p (or o) conjugates directly with the charge (σ^+(p-OMe) = –0.78) while at m only its –I effect operates (σ^+(m-OMe) = +0.05). A –M group at p (σ^+(p-NO_2) = +0.79) is worse than at m (+0.67), because at p it is conjugated with the charge-bearing carbon.
- **Halogens** have strong –I but a real +M effect (lone-pair donation); at p-F the two nearly cancel (σ^+ = –0.07), at p-Cl –I wins slightly (+0.11), at m-Cl only –I operates (+0.40).
- **Resonance requires coplanarity.** A π donor can only help if its p orbitals can align with the empty p orbital: see 4.8 (steric inhibition of resonance) and the vinyl/aryl cations (empty orbital in the nodal plane of the π system: no resonance).
- **Resonance energy** is large for the first donor and levels off for successive ones: allyl ≈ 20+ kcal mol^{-1} better than a localised secondary cation of the same substitution in the gas phase; each further vinyl unit adds less (pentadienyl, heptatrienyl).
- **The ranking of mesomeric donors** depends on orbital energy match and overlap: N (2p) > O (2p) > S (3p, poorer overlap but more polarisable) for the same carbon framework; in solution oxocarbenium ions are further stabilised by solvation.

4.3 Hyperconjugation

Donation of σ-bond electrons into the empty p orbital: **σ(C–H) → p** and **σ(C–C) → p** (and σ(C–M) → p for M = Si, Ge, Sn, which are much better donors because σ_{C-M} lies higher in energy). The stabilisation scales with (i) the donor ability of the σ bond (C–C ≳ C–H in the gas phase; C–Sn > C–Ge > C–Si ⩾ C–C), (ii) the number of aligned bonds and (iii) the

alignment angle θ between the σ bond and the empty p orbital (overlap ∝ $\cos^2\theta$; zero at 90°). In a freely rotating alkyl group on average ≈ 2 of the 3 C–H bonds are well aligned at any time; **simply counting α-hydrogens is not a valid ranking device** – t-Bu^+ (9 α-H) vs 2-adamantyl, bridgehead or conformationally locked cations show that orbital alignment, strain and polarisability also matter. Computations show that C–C hyperconjugation is as important as C–H; polarisability (stabilisation of the charge by larger alkyl groups) makes larger alkyl groups better in the gas phase (recalled HIA, kcal mol^{-1}: t-Bu^+ ≈ 231 → t-amyl$^+$ ≈ 225 → Et_3C^+ lower still).



C–H parallel to the p axis: maximum overlap

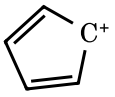
Figure 4.2 σ(C–H) → p hyperconjugation (the C–H bond must lie roughly parallel to the p-orbital axis).

Heteroatom hyperconjugation. Lone pairs and σ bonds of heteroatoms attached β to the cation can also donate: β-silicon (C–Si anti-periplanar to the leaving group) stabilises a β-carbocation by ≈ 38 kcal mol^{-1} relative to hydrogen in some computations (≈ 10^{12} rate acceleration in rigid systems), β-Ge and β-Sn are even stronger; β-halogens at 3° carbons may bridge instead (4.6). Negative hyperconjugation (n→σ*, e.g. in α-sulfonyl carbanions) is the anionic analogue and does not apply to cations.

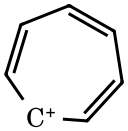
4.4 Aromaticity and antiaromaticity



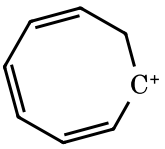
cyclopropenium $C_3H_3^+$ 2π, aromatic



cyclopentadienyl $C_5H_5^+$ 4π, antiaromatic



tropylium $C_7H_7^+$ 6π, aromatic



homotropylium $C_8H_9^+$ homoaromatic

Figure 4.3 Cyclic cations with a planar array of p orbitals: (4n+2) π electrons stabilise (cyclopropenium, tropylium); 4n π electrons destabilise (cyclopentadienyl cation, a triplet-ground-state species that is far less stable than the allyl cation). Homotropylium is stabilised by through-space (homo)conjugation across one sp^3 carbon.

- **Aromatic cations:** cyclopropenium (pK_{R+} of $C_3Ph_3^+$ +3.1; tris(dialkylamino)- and tricyclopropyl derivatives far higher), tropylium (pK_{R+} +4.7: isolable as stable Br^-, BF_4^- salts, planar, all C–C equal), cyclooctatetraene dications.
- **Antiaromatic cations:** cyclopentadienyl cation and the 9-fluorenyl cation destabilised by 4π delocalisation. The 9-phenylfluorenyl cation (pK_{R+} ≈ –10.8) is some 10^4 times less stable than trityl (–6.6) although the fluorenyl ring is planar – a clean demonstration of antiaromatic destabilisation. By contrast the 9-phenylxanthylium ion

(pK_{R+} ≈ +1.1) is strongly stabilised: the ring is planar and the oxygen lone pair donates without producing a 4π cycle.

- **Always check geometry and electron count** before declaring a cyclic conjugated cation aromatic: planarity may be unattainable (e.g. cyclooctatetraenyl cation puckers), and an sp^3 centre interrupts the cycle.

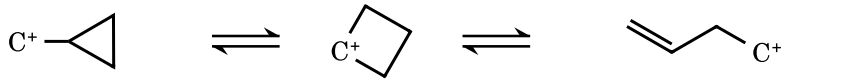
4.5 Cyclopropylcarbinyl stabilisation and 'dancing resonance'

A cyclopropane C–C bond is a bent ('banana') σ bond with high p character and high-lying HOMO; the cyclopropyl group therefore behaves like a π donor towards an adjacent empty p orbital – **provided the cation adopts the *bisected* conformation**, in which the empty p orbital is parallel to the plane of the ring and overlaps the two vicinal ring bonds. In the perpendicular conformation the overlap is nil. As a donor the cyclopropyl group is better than phenyl (σ^+: c-C_3H_5 –0.41, C_6H_5 –0.18) and, in tertiary systems, better than methyl: tricyclopropylcarbinyl cation (pK_{R+} –2.3) is more stable than trityl (–6.6).



bisected: p ∥ ring plane — good overlap perpendicular: poor overlap

Figure 4.4 Cyclopropylcarbinyl cation: bisected (left, strongly stabilised) and perpendicular (right, no stabilisation) conformations. Orbital view; rings viewed edge-on.



cyclopropylcarbinyl **cyclobutyl** **homoallyl (contributor)**

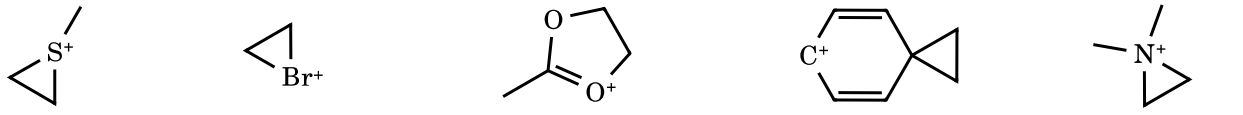
*Figure 4.5 The $C_4H_7^+$ manifold. The solvolysis of cyclopropylcarbinyl and of cyclobutyl derivatives gives, in water, cyclopropylcarbinol : cyclobutanol : homoallyl alcohol ≈ 48 : 47 : 5 (Roberts, Mazur). These are **product ratios**, not cation populations; the cation is better described as a single σ-delocalised 'bicyclobutonium'-like structure than as a mixture of three distinct classical ions.*

'Dancing resonance' describes how, in this ion, the three CH_2 carbons exchange identity by rapid, degenerate rearrangement through the delocalised structure, so that the carbons become equivalent on the NMR timescale. It is not ordinary resonance, because the nuclei move. The textbook drawing of three localised cations connected by ⇌ arrows is therefore a *caricature*: calculations and superacid NMR place the bicyclobutonium-like (σ-bridged, C_s) and the bisected cyclopropylcarbinyl structures within about 1 kcal mol^{-1} of each other and the barrier between them is minute; the open homoallyl and the classical puckered cyclobutyl ions are not distinct, deep minima.

4.6 Neighbouring-group participation and bridged ions

A genuine bridged intermediate forms when a group with a lone pair, π bond or σ bond **displaces the leaving group from the rear** (anti-periplanar) and ends up bonded to *both* carbons. It differs from ordinary resonance in three ways: (i) the neighbour **moves** toward the cationic carbon and a new σ bond (3-membered or 5-membered ring) is formed;

(ii) the stereochemical outcome is dictated by two successive inversions (net retention); (iii) the rate is higher than the intrinsic leaving-group ability suggests (anchimeric assistance).



episulfonium (S) **bromonium (Br)** **acetoxonium (O, 5-ring)** **phenonium (π, spiro)** **aziridinium (N)**

Figure 4.6 Bridged onium and phenonium ions. In each, positive charge is carried largely by the bridging heteroatom (or spread over the cyclohexadienyl ring in phenonium) and no open 'primary' cation exists.

- **Order of anchimeric power** for a given leaving group: roughly S ≈ N (free lone pair) > I > Br > Cl ≈ aryl > O (ether) – a lone pair donates best when its atom is large/polarisable and not strongly bound to its own nucleus; the actual rate enhancement depends on solvent, ring size formed and conformation.
- **π participation:** an aryl ring attacks C from the rear through its π face to give phenonium (Cram, Winstein); a C=C does the same in anti-7-norbornenyl tosylate (≈ 10^{11} faster than 7-norbornyl) and in homoallyl → cyclopropylcarbinyl systems.
- **σ participation:** a C–C or C–H σ bond (e.g. C1–C6 in exo-2-norbornyl; Wagner–Meerwein) acts as the nucleophile – see 4.9.
- **Test to separate bridging from resonance:** if two structures differ in the *position of atoms* (a ring closes), bridging is involved; if only electrons move, it is resonance.

4.7 d-Orbital resonance claims

Older textbooks explain the stabilisation of α-sulfonyl and α-phosphoryl carbanions, the planarity of silylamines, or the shortness of S–O bonds in sulfones/phosphates by **d_π–p_π bonding** (3d orbitals on S, P, Si accept electron density). Modern calculations show that 3d orbitals are too high in energy and too diffuse to participate significantly; d-functions in basis sets act as polarisation functions only. The effects are explained instead by: (i) **negative hyperconjugation** – donation of a lone pair into an antibonding σ*(S–O), σ*(S–C) or σ*(P–C) orbital; (ii) **polarisation** of the large, polarisable atom; (iii) strongly polarised 'ylidic' bonds (S^+–O^-, P^+–C^-) without a true double bond and with the octet intact. For **carbocations** the practical consequences are: α-silyl and α-sulfonyl substituents do *not* stabilise a cation (α-silyl cations are destabilised relative to alkyl; a sulfone is a strong –I group), whereas **β-silyl** stabilisation is σ-hyperconjugative and strong; and a sulfur lone pair (R–S–C^+) donates through *p–p* overlap (thionium ions) without d orbitals. Hypervalent descriptions (sulfonium ylide S=C) are therefore teaching shorthand, not electronic reality.

4.8 Steric inhibition of resonance (SIR)

If bulky ortho (or flanking) groups force the π donor out of coplanarity with the cationic centre, overlap falls as cos θ (energy as $\cos^2\theta$) and the resonance stabilisation is partly lost. The classic quantitative case: the dimethylamino group is an extreme π donor (σ^+ = −1.7) in 4-(dimethylamino)benzyl and trityl-type cations; but when two methyl groups flank the NMe_2 (3,5-dimethyl-4-dimethylamino) the amine is twisted ≈ 60–90° out of the

ring plane, the n→π donation is largely switched off and the group behaves close to a simple inductive substituent (σ^+ much less negative; the nitrogen lone pair cannot reach the charge).

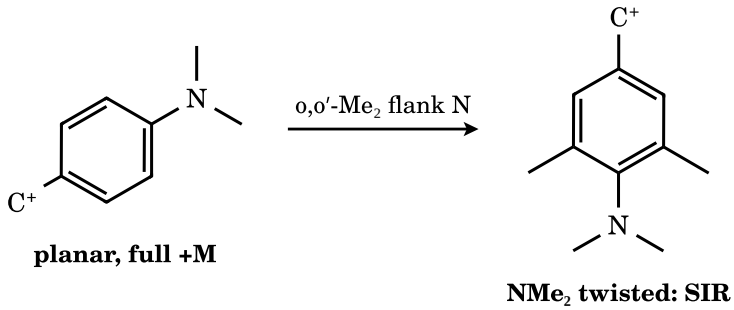


Figure 4.7 Steric inhibition of resonance: the flanking methyls (3,5 to the CH_2^+, ortho to N) twist the amine; the nitrogen lone pair then lies in the ring plane and cannot overlap with the π system.

> **No ranking by sterics alone.** SIR changes the *resonance* contribution only. Methyl groups at the same time exert +I/hyperconjugation and (in planar systems) can stabilise. Comparisons in Part 5 therefore always fix the other effects or state the observable.

4.9 Nonclassical carbocations

In a **classical** carbocation the charge is localised on one carbon (or delocalised by resonance only) and every bond is a 2c–2e bond. In a **nonclassical** ion σ electrons are delocalised over three or more nuclei by a 3c–2e bond, with *partial* bonds drawn as dotted lines. The best-established examples: 2-norbornyl, cyclopropylcarbinyl/bicyclobutonium, phenonium (aryl-bridged), protonated cyclopropane ($C_3H_7^+$ corner-protonated form), halonium ions, and the carbonium ions CH_5^+, $C_2H_7^+$.

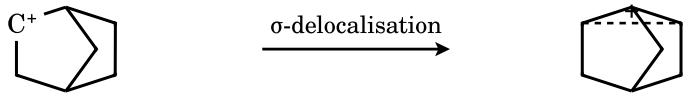


*Figure 4.8 2-Norbornyl cation: classical structure (left, positive charge on C-2) and the σ-bridged non-classical structure (right, schematic: the C1–C6 σ bond delocalises toward C-2 – dotted line – and the positive charge is shared by C1, C2 and C6 inside the three-centre triangle). The two are **not** resonance forms (atoms move); the bridged structure is the better description of the real ion.*

- **Kinetic evidence** (Winstein, 1950s): exo-2-norbornyl brosylate solvolyses ≈ 350 times faster than the endo isomer; both give exo product only and the product from optically active exo is racemic, as expected if σ(C1–C6) assists ionisation anti-periplanar to the exo leaving group, producing a symmetric (C_s) bridged ion that is attacked at either of two equivalent carbons.
- **Spectroscopic evidence** (Olah, 1960s–): in superacid at low temperature the ^{13}C NMR shows rapid 6,2- and 3,2-hydride shifts and Wagner–Meerwein exchange that average carbon signals; at very low temperature (–150 °C, solid-state studies) the signals match a bridged structure with a strongly deshielded pentacoordinate-type carbon environment.
- **Structural evidence:** a crystal structure of the 2-norbornyl cation with an $[Al_2Br_7]^-$ counterion (Scholz, Kramer, Sieber et al., *Science* 2013) shows the bridged geometry, with C–C distances between the three bridging carbons intermediate between single

and bridged-cyclopropane values. (Recalled from the literature; the source was not accessible here.)

- **The classical side (H. C. Brown):** the high exo/endo ratio can be explained by steric hindrance to ionisation of the endo isomer and by rapid equilibration of classical ions. The consensus today is that the **parent 2-norbornyl cation is bridged**, whereas tertiary derivatives (2-methyl-2-norbornyl, 2-aryl-2-norbornyl) are **classical** because the substituent already supplies enough stabilisation.
- **General rule:** bridging is a 'last resort' stabilisation. It occurs when the classical ion would be primary or secondary and unstabilised and the geometry offers an anti-periplanar σ (or π, or lone-pair) donor; it disappears when an aryl, cyclopropyl or tertiary substitution stabilises the cation already.

4.10 Carbonium ions (hypercoordinate carbon)

CH_5^+ (Figure 1.4) is the prototype. It has **eight** valence electrons for **five** C–H connections, so at least one bond is 3c–2e; the structure is fluxional (all five H exchange) and has an H_2 unit loosely bound to CH_3^+. $C_2H_7^+$ (protonated ethane) has a C–C or C–H σ bond protonated. In superacid, alkanes are protonated to carbonium ions, which lose H_2 (→ carbenium) or CH_4/alkanes (→ smaller carbenium ions: protolytic cracking, isomerisation). **Carbonium and carbenium ions are different bonding types and cannot be put on one stability scale by counting substituents.** Their relation is chemical, not energetic ranking: $R_3C{-}H + H^+ \rightleftharpoons [R_3C(H_2)]^+ \rightarrow R_3C^+ + H_2$.

4.11 Additional effects

- **Ring strain and I-strain.** Ionisation changes carbon from sp^3 to sp^2. H. C. Brown: 1-methylcyclopentyl chloride solvolyses ≈ 40× faster than t-BuCl, 1-methylcyclohexyl ≈ 3× slower (the six-membered ring loses a stable staggered arrangement and gains eclipsing strain), 1-methylcycloheptyl faster again; cyclopropyl and cyclobutyl cations are strained (bond angle 60°/90° versus 120°) and ionise only with ring-opening/participation.
- **Bridgehead constraint.** The cation must be able to flatten. Rates of solvolysis (80 % EtOH, 25 °C, relative to t-BuBr): 1-adamantyl ≈ 10^{-3}, bicyclo[2.2.2]oct-1-yl smaller, bicyclo[2.2.1]hept-1-yl (norbornyl) ≈ 10^{-13} (orders of magnitude recalled from Bingham–Schleyer; the sequence is secure) – a measure of the strain, not just of the cation's intrinsic stability (Bartlett, Schleyer).
- **Geometry/alignment.** Anti-periplanar σ bonds (C–Si, C–C) and gauche vs anti hyperconjugation; bisected vs perpendicular cyclopropylcarbinyl; rigid frameworks fix the alignment.
- **Field effects** of remote polar groups; **counterion** and ion pairing (a tight, nucleophilic counterion such as Cl^- can covalently collapse the cation; SbF_6^- cannot); **solvation** (H-bond donors stabilise the incipient anion; ion-dipole solvation favours small, localised cations, which is why alkyl order inverts in solution relative to the gas phase).
- **Isotope effects.** A CD_3 group is a poorer hyperconjugative donor than CH_3 (zero-point-energy difference): β-d_3 substitution slows ionisation by a factor ≈ 1.3–2.4 per CD_3, a probe of hyperconjugation in the transition state.

Effect	Acts through	Strength guide	Common error to avoid

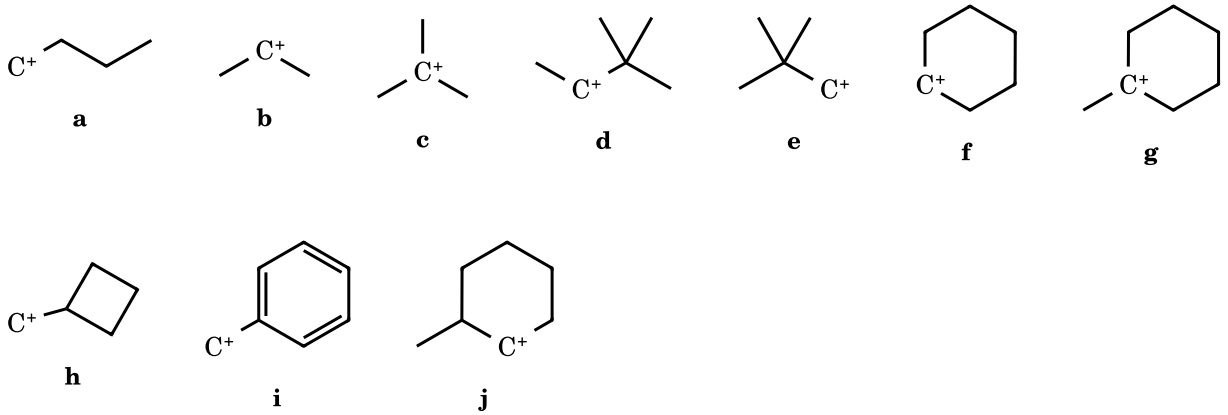
Resonance (+M)	p(π)/n → p	NR_2 > OR > SR > aryl > vinyl; strongest, needs coplanarity	forgetting SIR; counting meta as conjugated
Hyperconjugation	σ → p	C–Sn > C–Ge > C–Si > C–C ≳ C–H; angle-dependent	counting α-H blindly
Induction/field	σ framework, space	–I destabilises; +I stabilises; falls ≈ 2–3× per bond	overlooking +M of α-heteroatoms
Aromaticity	cyclic (4n+2) π	very large (tropylium, cyclopropenium)	calling 4n cations 'stable'
Ring strain	rehybridisation	I-strain (5 > 6), bridgehead penalties	ignoring conformational change on ionisation
Bridging (σ/π/n)	3c–2e or 3-ring onium	large for secondary/primary systems only	drawing both bridged and open ions as separate intermediates
Solvent/ion pair	electrostatics, H-bonding	can invert gas-phase orders	using solution rates as intrinsic stability

Part 7 Carbocation Rearrangements — Counting Problems (opening section)

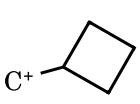
Counting problems test whether you can *recognise* a rearrangement before drawing any mechanism. Rules used (JEE-Advanced level): a carbocation rearranges by a 1,2-H, 1,2-alkyl, 1,2-aryl or ring-bond shift **only if the new cation is more stable** (more substituted, conjugated, or less strained) or the new ion is trapped irreversibly; shifts to an equally stable cation (secondary → secondary, tertiary → tertiary) are degenerate or unproductive and are **not** counted as 'rearranging'; tertiary, benzylic and allylic cations do not rearrange to a less stable ion. Primary cations are usually not true intermediates: the shift is concerted with the departure of the leaving group, but for counting purposes such a shift is 'a rearrangement'. The answers below were cross-checked with a small shift-enumeration program (every 1,2-shift of every cation was generated and the product classified) and by hand.

7.1 How many rearrange? How many ring-expand? How many shifts are possible?

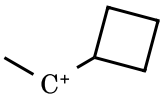
R1. How many of the following carbocations (a–j) will **undergo rearrangement** by a 1,2-shift to give a more stable carbocation? (Integer answer)



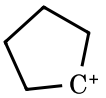
R2. How many of the carbocations (a–h) can undergo **ring expansion** by a single 1,2-shift to a more stable cation (more substituted and/or less strained)? (Integer answer)



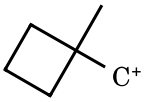
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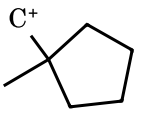
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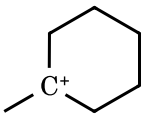
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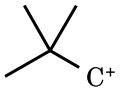
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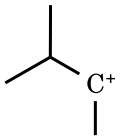


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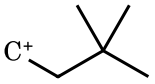
h

R3. The 3-methylbutan-2-yl cation, CH_3–CH^+–$CH(CH_3)_2$, can undergo different 1,2-shifts. (i) How many **different** carbocations (other than the starting ion itself) can be formed by one 1,2-shift of H or CH_3? (ii) How many of these are **more stable** than the starting ion? Give (i), (ii).



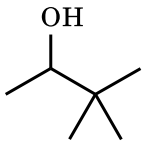
start

R4. How many 1,2-shifts, at minimum, convert the primary 3,3-dimethylbutyl cation, $(CH_3)_3C$–CH_2–CH_2^+, into the **most stable** carbocation that is reachable? (Integer answer; each hydride, alkyl or ring-bond shift counts one.)



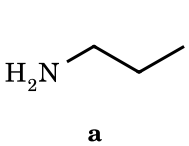
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R5. 3,3-Dimethylbutan-2-ol is heated with conc. H_3PO_4 (E1). How many **different alkenes** (structural isomers, counting only constitutional isomers, no stereoisomers arise) can be formed from *all* the carbocations that are intermediates, including the rearranged one? (Integer answer)

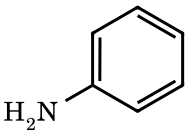


alcohol

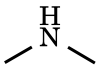
R6. Each compound (a–h) is treated with $NaNO_2$/HCl at 0–5 °C, then the solution is allowed to stand. **How many** of the compounds give a **carbocation** as an intermediate (free, or as a tight ion pair, formed by loss of N_2)? (Integer answer)



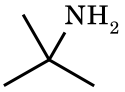
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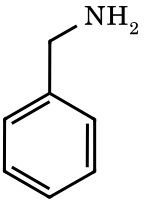
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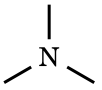
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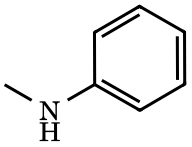
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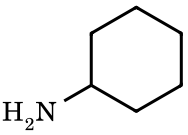
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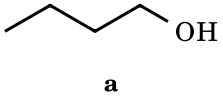


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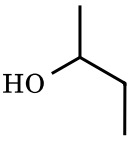


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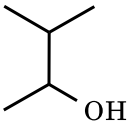
R7. Each alcohol (a–h) is heated with conc. HBr. **How many** give the alkyl bromide (or major product) with a **rearranged carbon skeleton or rearranged position of Br** through a carbocation or a concerted shift? (Integer answer)



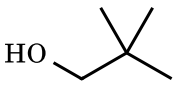
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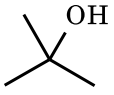
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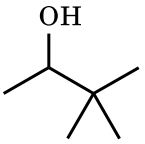
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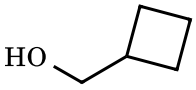
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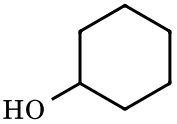
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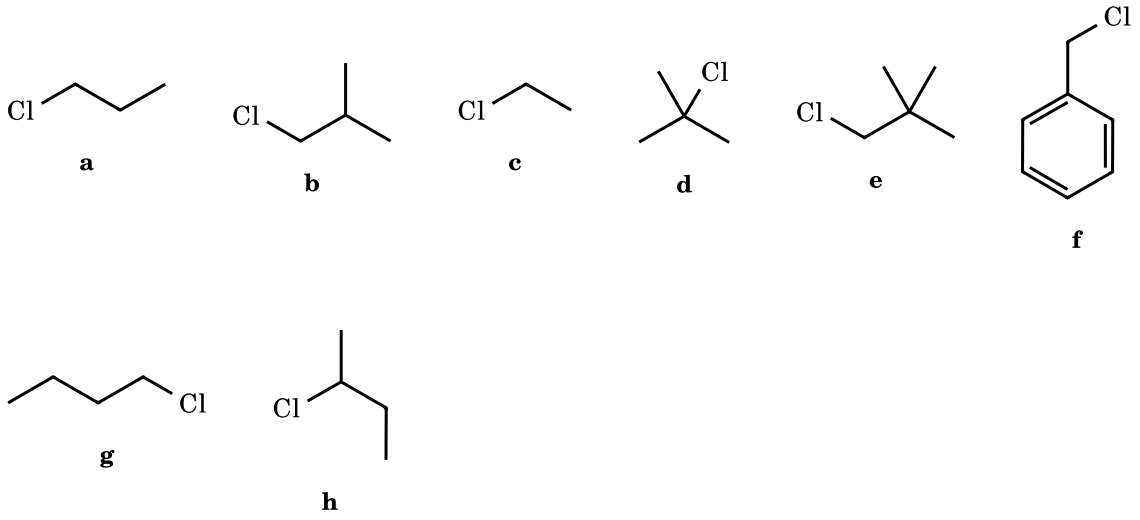


g



h

R8. Benzene is alkylated with each alkyl chloride (a–h) and $AlCl_3$. **How many** give a product in which the carbon skeleton or the point of attachment of the alkyl group **differs from the alkyl chloride used** (a rearranged product as the major product)? (Integer answer)



Answers Part 3 — Leaving-Group Problems: Answers and Explanations

LG1.

Answer: D > C > B > A

Rate-limiting heterolysis of C–X. Bond strength (C–I 57 ≪ C–F 110 kcal mol^{-1}), polarisability and conjugate-acid strength (HI −10, HBr −9, HCl −7, HF +3.2) all point the same way. Solvation of F^- by hydrogen bonding is the strongest of the four but cannot make up for the C–F bond: t-BuF is usually considered inert to neutral solvolysis. The three heavier halides differ only by modest factors (a few-fold between I and Br; tens between Br and Cl) because the TS is highly ionic and the solvent levels the differences.

LG2.

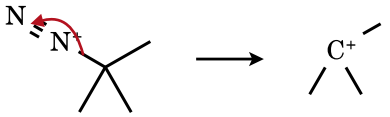
Answer: A > B > C > D

Leaving-group quality follows conjugate-acid strength: CF_3SO_3H (≈ −14) ⩾ TsOH (≈ −2.8) ⩾ CH_3CO_2H (4.76) ⩾ H_2O (15.7). Triflates ionise ≈ 10^4–10^8 times faster than tosylates depending on substrate; acetates are practically inert under these conditions, and the alcohol **D** does not ionise at all (HO^- is not a leaving group without protonation). A secondary substrate is chosen so that all three active derivatives react by borderline S_N1/S_N2 with solvent assistance; the order is the same for either.

LG3.

Answer: A > B > C > D

N_2 is a vastly better leaving group than anything else (a very stable, non-basic molecule; ionisation of an alkyl diazonium is practically barrierless). H_2O leaves from the protonated alcohol readily at 25 °C in acid (conjugate acid H_3O^+, pK_a −1.7). NH_3 is a much stronger base (NH_4^+ pK_a 9.25), so the t-butylammonium ion is stable at room temperature and needs heat or very strong acid to lose NH_3. Hydroxide in **D** is the worst: C–OH ionisation does not occur without protonation (D is therefore last, essentially inert).



LG4.

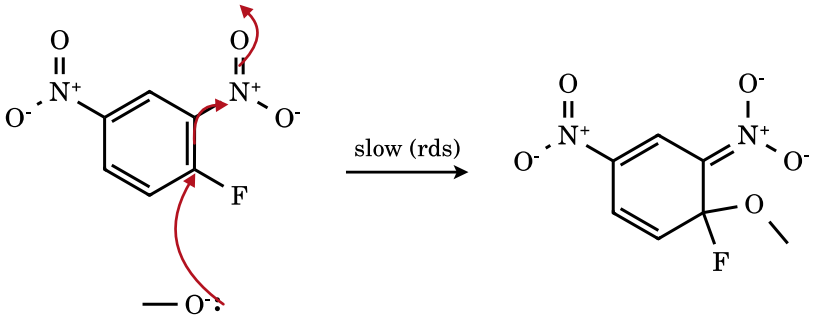
Answer: C > D > B > A

Carboxylate leaving groups follow the pK_a of the parent acid: CF_3CO_2H 0.5 < p-$O_2NC_6H_4CO_2H$ 3.44 < $C_6H_5CO_2H$ 4.20 < CH_3CO_2H 4.76. The same trend is the basis of the classical use of p-nitrobenzoates as 'better' leaving groups than acetates in mechanistic work on tertiary benzylic systems. Resonance donation within the ester (which strengthens the C–O bond) is similar for all four and cancels.

LG5.

Answer: A > B > C

In SNAr the rate-determining step is *addition* of the nucleophile to give the anionic Meisenheimer complex; C–X cleavage comes afterwards (fast). Fluorine, the most electronegative halogen, lowers the energy of the first TS by induction and polarises C–X; the C–F bond strength is irrelevant to the rate. Hence the 'element effect' F ⩾ Cl ≳ Br > I, the reverse of S_N2 – a decisive demonstration that leaving-group ability is mechanism-dependent. (Cl and Br are close; iodine is slowest.)



LG6.

Answer: D > C > B > A

The poorer the leaving group, the less C–LG bond cleavage has occurred in the E2 transition state; the TS becomes carbanion-like (E1cB-like) and the base removes the more acidic, more accessible hydrogen from the CH_3 group, giving the Hofmann alkene. Literature trend for 2-pentyl X with ethoxide: I < Br (≈ 30 %) ≲ Cl (≈ 33 %) < F (≈ 70 %) < NMe_3^+ (≈ 95 % 1-pentene); I and Br differ only slightly. Here LG ability controls *selectivity*, not only rate.

LG7.

Answer: A > B > C > D > E

Y (recalled values): H_2O 3.49 > HCO_2H 2.05 > CF_3CH_2OH 1.15 > CH_3OH −1.09 > C_2H_5OH −2.03. Ionising power combines hydrogen-bond donation to the departing chloride (high for water, HCO_2H, fluorinated alcohols) with dielectric screening and low nucleophilicity. TFE is a poor nucleophile yet an excellent ionising medium, so S_N1 reactions of secondary substrates are favoured there; a leaving group's 'ability' is therefore also a property of the solvent.

LG8.

Answer: A > C > B

E2 needs the C–H and C–Br bonds anti-periplanar, which on a chair means *both axial*. In the cis isomer (A) the Br is axial (the locked t-Bu is equatorial), so reaction is fast. In the trans isomer (B) Br is equatorial, no anti-periplanar H exists, the ring would have to flip into the diaxial-t-Bu chair, and E2 is slow (substitution/E1 intrude). Bromocyclohexane (C) populates the axial conformer to a modest extent (≈ 30 %), so its rate lies between. The axial/equatorial labels refer to chairs, which these flat drawings cannot show; wedge/dash give the relative (cis/trans) configuration.

LG9.

Answer: A > B > C > D

The rate follows both the leaving-group stability (Cl^-, $pK_a(HCl)$ −7; $CH_3CO_2^-$ 4.76; EtO^- 16; NH_2^- 38) and the ability of the substituent to donate into C=O (N ⩾ O > Cl). The tetrahedral intermediate expels the best leaving group; amides, with strong n(N) → π*(C=O) donation and a very poor LG, are the least reactive. Contrast with S_N1/S_N2 at sp^3 carbon, where resonance donation is irrelevant.

LG10.

Answer: A > B > C

Trimethyloxonium (Meerwein) salts are among the strongest methylating agents; trimethylsulfonium is a weak methylating agent (it needs warming), and tetramethylammonium is inert. The leaving groups are Me_2O, Me_2S and Me_3N. Basicity (pK_{aH} of Me_3N 9.8 ⩾ Me_2O ≈ −3.8, Me_2S ≈ −6 to −7) explains why ammonium is last, but *not* why oxonium beats sulfonium: sulfur is the weaker base. Here the intrinsic C–X^+ bond strength and the charge density at the cationic centre dominate – a clear failure of a pure pK_a rule. (Experimental trend: O^+ ⩾ S^+ ⩾ N^+.)

LG11.

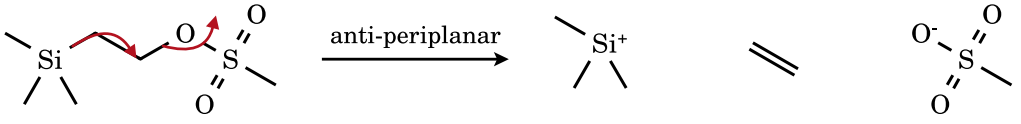
Answer: A > B > C > D

Ag^+ coordinates the halogen lone pair and 'pulls' the bromide off (halophilic assistance); the substrate then ionises. The ease of forming the cation is 3° ⩾ 2° > 1° (t-Bu^+ ⩾ i-Pr^+ ⩾ n-Bu^+); the n-butyl bromide reacts only slowly (a partial S_N2-like displacement by solvent) and bromobenzene (an sp^2 aryl cation would be needed) gives no precipitate. Same leaving group, different cation – the LG is only as good as the cation it can generate.

LG12.

Answer: A > B > C

β-Silicon effect: a C–Si σ bond is an excellent hyperconjugative donor (σ_{C-Si} is higher in energy than $\sigma_{C-C}/\sigma_{C-H}$) and stabilises a β cation by σ→p donation (a bridged, silicon-stabilised cation). The overlap is best when C–Si and C–OTs are **anti-periplanar**, which the trans isomer reaches in the diaxial chair (A: very large acceleration, ≈ 10^{12} in rigid models). The cis isomer (B) has them gauche (synclinal) only: still accelerated (≈ 10^4 in rigid models) but far less. Cyclohexyl tosylate (C) has no silicon. (Orders of magnitude quoted from Lambert's work, recalled.)



LG13.
Answer: A > B > C > D
The aryl ring attacks the carbon carrying OTs from the rear, forming a bridged phenonium (spiro-cyclohexadienyl) ion; the rate rises with the electron-donating power of the para substituent (σ^+: OMe −0.78, Me −0.31, H 0, NO_2 +0.79). The p-nitro compound cannot use the ring and reacts by the slower, unassisted S_N2-like route (k_s). The 'leaving group' is identical (TsO^-) in all four; the substituent changes how effectively it is displaced.

LG14.
Answer: A > B > C > D
The leaving group is the protonated oxygen (an alcohol or ether oxygen that becomes neutral on C–O cleavage), and the driving force is the release of ring strain: oxirane (≈ 27 kcal mol^{-1}) and oxetane (≈ 25) are strained; oxolane (THF, ≈ 6) is slightly strained; oxane (THP, ≈ 0.5) is not. Epoxides react far faster than oxetanes despite similar strain energies because the TS gains from poorer C–O–C orbital overlap (bent bonds) in the three-membered ring; THF is cleaved only under more forcing conditions, THP hardly at all.

LG15.
Answer: A > B > C > D
The sulfonate anion is stabilised by electron-withdrawing substituents (lower pK_a of $ArSO_3H$), so the leaving-group ability follows the Hammett σ constants: NO_2 (σ_p +0.78) > Br (+0.23) > CH_3 (−0.17) > OCH_3 (−0.27); ρ for the leaving group is positive and modest. Brosylates and nosylates are therefore used when tosylates are too unreactive; mesylates are comparable to tosylates.

Answers Part 7.1 — Counting Problems: Answers and Explanations

R1.
Answer: 5 (a, d, e, h, j)

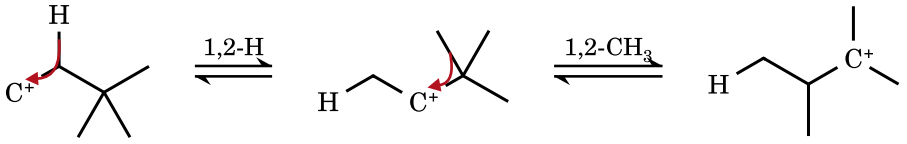
Item	?	Reason
a n-butyl	**Yes**	primary; 1,2-H shift → 2-butyl (secondary)
b isopropyl	**No**	H or CH_3 shift would give a primary cation, or a degenerate secondary one
c tert-butyl	**No**	any shift gives a primary cation
d 3,3-dimethylbut-2-yl	**Yes**	1,2-CH_3 shift → tertiary 2,3-dimethylbut-2-yl
e neopentyl	**Yes**	1,2-CH_3 shift → tertiary tert-amyl
f cyclohexyl	**No**	H shift gives another secondary ring cation (degenerate); the ring does not strain-relieve
g 1-methylcyclohexyl	**No**	tertiary; shifts lead to less stable ions
h cyclobutylmethyl	**Yes**	ring expansion → cyclopentyl (secondary, strain relief), or H shift → 1-methylcyclobutyl (tertiary)
i benzyl	**No**	resonance-stabilised; a shift would give an unstabilised ion
j 2-methylcyclohexyl	**Yes**	1,2-H shift from the ring C–H bearing CH_3 → tertiary 1-methylcyclohexyl

R2.
Answer: 4 (a, b, d, e)

Item	?	Reason
a cyclobutylmethyl	**Yes**	ring C–C bond migrates to the CH_2^+: 4 → 5 ring, primary → secondary
b 1-cyclobutylethyl	**Yes**	ring bond migrates to the exocyclic cationic carbon: 2-methylcyclopentyl (secondary, ring strain relieved)
c cyclopentyl	**No**	the cation is in the ring; no exocyclic carbon to take a ring bond
d (1-methylcyclobutyl)methyl	**Yes**	ring bond migrates → 1-methylcyclopentyl, a tertiary cation with relief of strain (much better than the methyl-shift alternative)
e (1-methylcyclopentyl)methyl	**Yes**	ring bond migrates → 1-methylcyclohexyl (tertiary, strain-free); the competing methyl shift gives 1-ethylcyclopentyl (also tertiary), so ring expansion is the thermodynamic choice
f 1-methylcyclohexyl	**No**	tertiary cation on the ring: expansion impossible
g cyclobutyl	**No**	cation is in the ring (it can ring-contract to cyclopropylcarbinyl, the reverse)
h neopentyl	**No**	no ring: methyl shift is a rearrangement but not a ring expansion

R3.
Answer: (i) 2 (ii) 1
Shifts available: **H from C-3** → 2-methylbut-2-yl, tertiary (more stable); **CH_3 from C-3** → regenerates a 3-methylbutan-2-yl cation (degenerate, not a new ion); **H from the CH_3 on C-2** → 3-methylbutyl, primary (less stable). Distinct new cations: tertiary and primary = 2; more stable: only the tertiary = 1. (Hydride shift is the productive path, reached through a three-centre transition state with σ(C–H) → p overlap.)

R4.
Answer: 2
Step 1: 1,2-hydride shift from the CH_2 next to the quaternary carbon → 3,3-dimethylbut-2-yl (secondary). Step 2: 1,2-methyl shift from the quaternary carbon → 2,3-dimethylbut-2-yl (tertiary, the most stable ion in this manifold). A direct primary → tertiary shift is impossible because the tertiary centre is two bonds away. (The same cation is the one formed from 3,3-dimethylbutan-2-ol; see R5.)



R5.
Answer: 3
Unrearranged secondary cation (3,3-dimethylbut-2-yl): loses H^+ only from CH_3 (the other neighbour is quaternary) → **3,3-dimethylbut-1-ene** (minor). After the 1,2-CH_3 shift the tertiary 2,3-dimethylbut-2-yl cation loses H^+ from the CH (→ **2,3-dimethylbut-2-ene**, tetrasubstituted, major) or from a CH_3 (→ **2,3-dimethylbut-1-ene**). Total 3; the Zaitsev (2,3-dimethylbut-2-ene) dominates.

R6.
Answer: 4 (a, d, e, h)

Item	?	Reason
a n-propylamine	**Yes**	$R–N_2^+$ loses N_2 (with 1,2-H shift) → n-propyl/isopropyl cation; mixture of propanols, propene, chloropropanes
b aniline	**No**	$ArN_2^+ Cl^-$ is stable at 0–5 °C (resonance, sp^2 C–N); the aryl cation forms only on warming
c dimethylamine	**No**	secondary amine → N-nitrosodimethylamine, $Me_2N–N=O$
d tert-butylamine	**Yes**	t-Bu–N_2^+ → tert-butyl cation
e benzylamine	**Yes**	$PhCH_2N_2^+$ → benzyl cation
f trimethylamine	**No**	tertiary amine: only the ammonium nitrite salt; no diazonium
g N-methylaniline	**No**	secondary arylamine → N-nitroso-N-methylaniline
h cyclohexylamine	**Yes**	c-$C_6H_{11}N_2^+$ → cyclohexyl cation (and cyclohexene, cyclohexanol, ring-contracted products)

R7.
Answer: 4 (c, d, f, g)

Item	?	Reason
a butan-1-ol	**No**	primary, unbranched: S_N2 → 1-bromobutane
b butan-2-ol	**No**	secondary cation (S_N1/S_N2 mixture); a shift would give a primary or an identical secondary ion
c 3-methylbutan-2-ol	**Yes**	H shift → tertiary → 2-bromo-2-methylbutane
d neopentyl alcohol	**Yes**	concerted CH_3 shift → 2-bromo-2-methylbutane
e tert-butyl alcohol	**No**	tertiary cation
f 3,3-dimethylbutan-2-ol	**Yes**	CH_3 shift → tertiary → 2-bromo-2,3-dimethylbutane
g cyclobutylmethanol	**Yes**	ring expansion / rearrangement → bromocyclopentane and 1-bromo-1-methylcyclobutane type products
h cyclohexanol	**No**	secondary ring cation; H shift is degenerate

R8.
Answer: 4 (a, b, e, g)

Item	?	Reason
a 1-chloropropane	**Yes**	H shift in the R–Cl···$AlCl_3$ complex → isopropylbenzene (cumene)
b 1-chloro-2-methylpropane	**Yes**	H shift → tert-butyl → tert-butylbenzene
c chloroethane	**No**	no more stable cation reachable (ethyl is the only possibility)
d tert-butyl chloride	**No**	tertiary cation, no shift
e 1-chloro-2,2-dimethylpropane	**Yes**	CH_3 shift → tert-amyl → (2-methylbutan-2-yl)benzene
f benzyl chloride	**No**	benzylic cation; gives diphenylmethane without rearrangement
g 1-chlorobutane	**Yes**	H shift → sec-butyl → sec-butylbenzene as the major component (n-butylbenzene is also formed; the ratio depends on temperature, roughly 2 : 1 in favour of sec-butyl at 0 °C – recalled value)
h 2-chlorobutane	**No**	secondary → secondary: the same sec-butyl product